

**Prediction of Censored Lifetimes under  
Proportional Hazard Rate Models Based on  
Progressive Type-II Censored Samples: A  
Comparative Study on Exponential, Lindley,  
and Rayleigh Distributions**



*A project submitted in fulfillment of the requirements for the  
degree of Master of Science in Statistics*

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## CERTIFICATION

This is to certify that the project work entitled “**Prediction of Censored Lifetimes under Proportional Hazard Rate Models Based on Progressive Type-II Censored Samples: A Comparative Study on Exponential, Lindley, and Rayleigh Distributions**” has been prepared by **Diparna Santra** under my supervision and guidance. The dissertation is her original work, the result of careful research and investigation. The work of the dissertation is of the standard expected of a candidate for the M.Sc. Degree (Statistics) Program, and I recommend that it be sent for evaluation.

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**Abstract.** In this project, we discuss predictors of time-to-failure for units censored at multiple stages in a progressively censored sample from proportional-hazard rate models. The maximum likelihood, best unbiased, and conditional median predictors are considered. We also consider Bayesian point predictors for the times to failure of units. The comparisons are made among the exponential, Lindley, and Rayleigh base hazard rate models. A Monte Carlo simulation study is presented to illustrate all the prediction methods discussed in this project. Predictive estimates of censored lifetimes are presented for synthetic and real-life datasets for illustrative purposes.

*Key words and phrases:* Bayesian point predictor, best unbiased predictor, conditional median predictor, maximum likelihood predictor, Monte Carlo simulation, progressive Type-II censoring, proportional hazard rate model.

# 1 Introduction

Progressive Type-II censoring is widely used in reliability and survival experiments. Under this censoring mechanism, some of the surviving test units may be intentionally withdrawn at predetermined failure times during the experiment, thereby reducing the overall duration and cost of the life-testing procedure.

The conventional Type-I and Type-II censoring schemes are commonly employed in life-testing experiments. However, these classical censoring methods do not allow the withdrawal of experimental units before the test terminates. In many practical situations, such flexibility is necessary. For example, in wear or ageing studies, some units may need to be dismantled at intermediate stages in order to examine the deterioration mechanism in detail. Likewise, when the test items are costly or difficult to procure, removing a portion of the surviving units during the experiment allows them to be utilised in other investigations. Such practical considerations motivate the adoption of progressive censoring schemes, which offer greater flexibility and efficiency in reliability experiments.

The progressive Type-II censoring mechanism may be explained as follows. Consider that  $n$  experimental units are initially placed under a life-testing study. After the occurrence of the first failure,  $R_1$  surviving units are randomly withdrawn from the experiment. Similarly, immediately after the second observed failure, another  $R_2$  surviving units are randomly removed. This withdrawal procedure continues until the  $m$ -th failure, at which point all remaining surviving units are removed from the test. Consequently, the number of units withdrawn at the final stage is given by

$$R_m = n - m - \sum_{i=1}^{m-1} R_i.$$

The censoring numbers  $R_1, R_2, \dots, R_m$  are predetermined before the experiment begins. When

$$R_1 = R_2 = \dots = R_m = 0,$$

the experiment reduces to a complete sample case with  $n = m$ . On the other hand, if

$$R_1 = R_2 = \dots = R_{m-1} = 0,$$

then

$$R_m = n - m,$$

which represents the ordinary Type-II right censoring scheme. A comprehensive discussion on progressive censoring schemes and their statistical inference procedures can be found in Balakrishnan and Aggarwala (2000). A pictorial view is shown in Figure 1 to facilitate easy, comprehensive understanding for common readers.

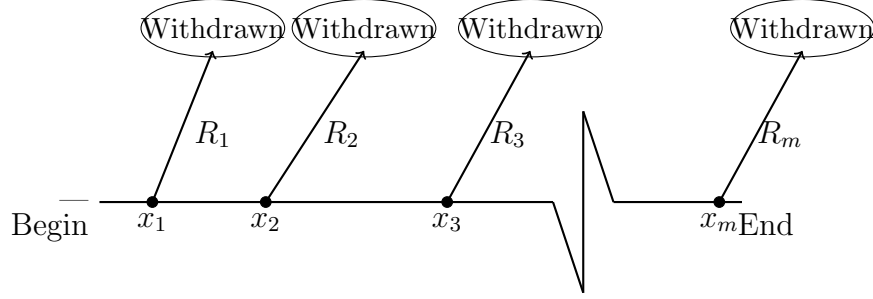


Figure 1: Progressive Type-II censoring scheme.

The prediction of censored or unobserved lifetimes has attracted considerable attention across actuarial studies, medical research, and reliability engineering. Several authors have contributed to the development of prediction methods under progressively censored sampling schemes. Viveros and Balakrishnan (1994) proposed conditional prediction intervals for future observations based on progressively Type-II-censored samples. Later, Balakrishnan and Lin (2002) derived exact prediction intervals for the final censored failure times of exponentially distributed lifetimes using the best linear unbiased estimator (BLUE). Basak et al. (2006) conducted an extensive study of point prediction techniques for censored failure times arising from progressively Type-II right-censored samples. More recently, Raqab et al. (2010) investigated various prediction procedures for failure times of units censored at different stages under a progressive censoring framework for the Pareto Type-II (Lomax) distribution.

Consider a baseline cumulative distribution function (cdf)  $F_0(\cdot)$  with corresponding hazard rate function  $r_0(\cdot)$ . A class of distributions having hazard function of the form  $\theta r_0(\cdot)$ , where  $\theta > 0$ , is referred to as the proportional hazard rate (PHR) family. In this setup,  $F_0(\cdot)$  serves as the baseline distribution function. Hence, if the random variable  $X$  belongs to the PHR family

associated with the baseline cdf  $F_0(\cdot)$ , then its distribution function can be expressed as

$$F(x; \theta) = 1 - [\bar{F}_0(x)]^\theta, \quad x \in B, \theta > 0, \quad (1)$$

where

$$\bar{F}_0(x) = 1 - F_0(x)$$

denotes the baseline survival function, and  $B$  represents the support of the baseline distribution. It may be observed that the baseline model itself is obtained when  $\theta = 1$ . The proportional hazard framework was first introduced by Cox (1972) and has since received considerable attention in statistical research. Several important lifetime distributions arise as special cases of this family, including the exponential, Pareto Type I, Pareto Type II, Beta, and Burr Type XII distributions. Further developments and applications of the PHR model may be found in Ahmadi et al. (2008, 2009) and Asgharzadeh and Valiollahi (2009, 2010a).

In the proportional hazards rate framework proposed by Cox (1972), the parameter  $\theta$  is typically treated as a function of explanatory variables, or covariates. Let  $z = (z_1, z_2, \dots, z_k)$  denote the vector of covariates. Then,  $\theta$  may be represented as a function of  $z$ , say  $\theta(z)$ , and the corresponding hazard function takes the form

$$r(x | \theta(z)) = r(x)\theta(z). \quad (2)$$

Different specifications for the covariate function  $\theta(z)$  may be considered in practice. Two widely used forms are the linear model

$$\theta(z) = \beta z, \quad (3)$$

and the log-linear model

$$\theta(z) = \exp(\beta z), \quad (4)$$

where  $\beta$  may represent a scalar or vector of regression coefficients. In particular, when the covariate function follows the log-linear form  $\theta(z) = \exp(\beta z)$ , the resulting proportional hazards model is known as the Cox regression model. Alternative functional forms for  $\theta(z)$  have also been explored in the literature. Additional discussions on these models may be found in Lawless (2003) and Marshall and Olkin (2007).



Corresponding to the distribution function in model (1), the associated probability density function can be written as

$$f(x; \theta) = \theta f_0(x) [\bar{F}_0(x)]^{\theta-1}, \quad -\infty \leq c < x < d \leq \infty, \quad \theta > 0, \quad (5)$$

where  $f_0(x)$  and  $\bar{F}_0(x)$  denote the baseline density and survival functions, respectively. Suppose that

$$X_{1:m:n}, X_{2:m:n}, \dots, X_{m:m:n}$$

represent the progressively Type-II censored order statistics obtained from a sample of size  $n$  under the censoring scheme  $(R_1, R_2, \dots, R_m)$  from the PHR model defined in (1). For notational convenience, the order statistics  $X_{i:m:n}$  will subsequently be denoted simply by  $X_i$ . The primary objective of this study is to investigate the prediction of lifetimes for all censored observations across different censoring stages. Specifically, attention is focused on predicting

$$Y = X_{i,(s)}, \quad s = 1, 2, \dots, R_i, \quad i = 1, 2, \dots, m,$$

based on the observed progressively censored sample

$$X = (X_1, X_2, \dots, X_m).$$

Here,  $X_{i,(s)}$  denotes the  $s$ -th order statistic among the  $R_i$  units withdrawn at the  $i$ -th censoring stage. A schematic diagram is shown in Figure 2.

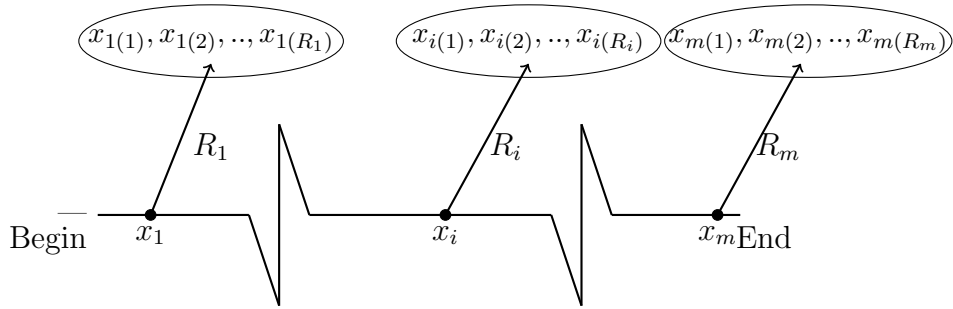


Figure 2: Progressive Type-II censoring scheme with Predictive lifetimes.

Asharzadeh and Valiollahi (2010b) studied point prediction for the proportional hazards family under progressive type-II censoring, focusing on the

exponential distribution. But the exponential distribution has its limitation of a constant failure rate. In the project, we compared estimators under the Rayleigh and Lindley distributions with those under the exponential distribution.

Section 2 is devoted to the development of several point prediction procedures under both classical and Bayesian frameworks. To demonstrate the applicability and performance of the proposed prediction techniques, a numerical illustration and a Monte Carlo simulation study are provided in Section 3.

## 2 Point Prediction

Let  $X_1, X_2, \dots, X_m$  denote a progressively Type-II right censored sample from model (1), with  $(R_1, R_2, \dots, R_m)$  being the progressive censoring scheme. Our interest is to predict  $Y = X_{i,(s)}$  ( $s = 1, 2, \dots, R_i$ ;  $i = 1, 2, \dots, m$ ) in all  $m$  stages of censoring based on the observed progressively Type-II right censored sample  $x = (x_1, \dots, x_m)$ .

In the project, we compare the estimators of  $\theta$ , arising out of three base distributions, such as standard exponential, standard Rayleigh, and standard Lindley, though the theory has been developed for the distributions involving parameter  $\lambda$ .

### • Exponential Distribution

The probability density function (pdf) of the exponential distribution with parameter  $\lambda > 0$  is

$$f(x; \lambda) = \lambda e^{-\lambda x}, \quad x > 0.$$

The survival function is

$$S(x) = e^{-\lambda x}.$$

Hence, the hazard rate function is

$$h(x) = \frac{f(x)}{S(x)} = \lambda.$$

- **Lindley Distribution**

The probability density function (pdf) of the Lindley distribution with parameter  $\lambda > 0$  is

$$f(x; \lambda) = \frac{\lambda^2}{\lambda + 1}(1 + x)e^{-\lambda x}, \quad x > 0.$$

The survival function is

$$S(x) = \left(1 + \frac{\lambda x}{\lambda + 1}\right) e^{-\lambda x}.$$

Thus, the hazard rate function is

$$h(x) = \frac{f(x)}{S(x)} = \frac{\lambda^2(1 + x)}{\lambda x + \lambda + 1} (< \lambda).$$

- **Rayleigh Distribution**

The probability density function (pdf) of the Rayleigh distribution with parameter  $\lambda > 0$  is

$$f(x; \lambda) = 2\lambda x e^{-\lambda x^2}, \quad x > 0.$$

The survival function is

$$S(x) = e^{-\lambda x^2}.$$

Therefore, the hazard rate function is

$$h(x) = \frac{f(x)}{S(x)} = 2\lambda x.$$

For the special case  $\lambda = 1$ , the hazard rate functions of the exponential, Lindley, and Rayleigh distributions reduce to

$$h_E(x) = 1,$$

$$h_L(x) = \frac{(1+x)}{(2+x)},$$

and

$$h_R(x) = 2x,$$

respectively. The exponential distribution exhibits a constant hazard rate over time, reflecting a memoryless failure mechanism. In contrast, the Lindley and Rayleigh distributions possess increasing hazard rate functions. Although the hazard rate of the Lindley distribution increases with time, its growth remains bounded and approaches a finite limit, equal to that of the exponential distribution. On the other hand, the Rayleigh hazard rate increases linearly without bound, providing greater flexibility in modelling ageing phenomena. Owing to these distinct hazard-rate characteristics, the exponential, Lindley, and Rayleigh distributions constitute a useful set of baseline models for comparative studies of different failure-rate behaviours.

## 2.1 Maximum Likelihood Predictor

Maximum likelihood prediction has been widely employed for forecasting future observations in a variety of statistical settings. Several studies, including those of Kaminsky and Rhodin (1985), Basak and Balakrishnan (2003), and Basak et al. (2006), have investigated the properties and applications of this approach. In the context of progressively censored samples, Basak et al. (2006) established the existence and uniqueness of the maximum likelihood predictor. They also examined its theoretical properties and derived conditions under which the predictor possesses desirable features such as unbiasedness and consistency.

The maximum likelihood prediction method extends the likelihood principle to simultaneously estimate unknown model parameters and predict future observations. Unlike Bayesian procedures, this approach does not require prior information and treats the prediction problem within a frequentist framework. Let

$$X = (X_1, X_2, \dots, X_m)$$

denote the observed sample and let  $Y$  represent an unobserved future random variable. Suppose that the joint probability density function of  $(X, Y)$  is given by

$$f(x, y; \theta),$$

where  $\theta$  is an unknown parameter. The objective is to infer the future value of  $Y$  using the information contained in the observed data  $X$ . To achieve this, the joint density is regarded as a likelihood function in both the unknown quantity  $y$  and the parameter  $\theta$ , and we define

$$L(y, \theta; x) = f(x, y; \theta) \quad (6)$$

to be the predictive likelihood function (PLF) of  $y$  and  $\theta$ .

Suppose  $Y_p = u(X)$  and  $\hat{\theta}_p = v(X)$  are statistics for which

$$L(u(x), v(x); x) = \sup_{(y, \theta)} L(y, \theta; x),$$

then  $u(X)$  is said to be the maximum likelihood predictor (MLP) of  $Y$  and  $v(X)$  the predictive maximum likelihood estimator (PMLE) of  $\theta$ . Note that the PLF can be rewritten as

$$L(y, \theta; x) = f(y|x; \theta)f(x; \theta).$$

So, when  $\theta$  is known, an MLP for  $Y$  is also a mode of the conditional distribution of  $Y$  given  $X = x$ .

Now, let  $X_1, X_2, \dots, X_m$  be a progressively Type II censored sample from model (1), with progressive censoring scheme  $(R_1, R_2, \dots, R_m)$ . Then, the joint density function of  $X = (X_1, X_2, \dots, X_m)$  is given (see Balakrishnan and Aggarwala (2000)) by

$$\begin{aligned} f(x; \theta) &= A \prod_{i=1}^m [f(x_i; \theta)[1 - F(x_i; \theta)]^{R_i}] \\ &= A \left[ \prod_{i=1}^m \frac{f_0(x_i)}{\bar{F}_0(x_i)} \right] \theta^m e^{-\theta T(x)}, \end{aligned} \quad (7)$$

where  $A = n(n-1-R_1)(n-2-R_1-R_2) \cdots (n-m+1-R_1 \cdots -R_{m-1})$  and  $T(x) = -\sum_{i=1}^m (R_i + 1) \ln \bar{F}_0(x_i)$ .

From (7), the maximum likelihood estimator (MLE) of  $\theta$  is derived to be

$$\hat{\theta}_{ML} = \frac{m}{T(x)}.$$

A useful property of progressively Type-II censored order statistics is that they possess a Markov structure. As a consequence, the conditional distribution of the censored observation  $X_{i,(s)}$  given the entire observed sample  $X$  depends only on the corresponding failure time  $X_i = x_i$  (see Balakrishnan and Aggarwala, 2000). Hence, conditioning on  $X = x$  is equivalent to conditioning on  $X_i = x_i$ .

Under this condition, the random variable  $X_{i,(s)}$  may be viewed as the  $s$ -th order statistic from a sample of size  $R_i$  drawn from the parent distribution truncated from the left at  $x_i$ . The corresponding truncated density is

$$\frac{f(y)}{1 - F(x_i)}, \quad y \geq x_i.$$

Therefore, if  $Y = X_{i,(s)}$  denotes the  $s$ -th ordered lifetime among the  $R_i$  units withdrawn at the  $i$ -th censoring stage, then the conditional density of  $Y$  given  $X_i = x_i$ , for  $y \geq x_i$ , can be expressed as

$$\begin{aligned} f(y|x_i; \theta) &= s \binom{R_i}{s} f(y; \theta) [F(y; \theta) - F(x_i; \theta)]^{s-1} \\ &\quad \times [1 - F(y; \theta)]^{R_i-s} [1 - F(x_i; \theta)]^{-R_i}. \end{aligned} \quad (8)$$

For model (1), (8) reduces to

$$\begin{aligned} f(y | x_i; \theta) &= s \binom{R_i}{s} \frac{f_0(y)}{\bar{F}_0(y)} [(\bar{F}_0(x_i))^\theta - (\bar{F}_0(y))^\theta]^{s-1} \\ &\quad \times [(\bar{F}_0(y))^\theta]^{R_i-s+1} [(\bar{F}_0(x_i))^\theta]^{-R_i}, \quad y \geq x_i. \end{aligned} \quad (9)$$

Consequently, the predictive likelihood function (PLF) of  $Y$  and  $\theta$ , can be rewritten as

$$\begin{aligned} L(y, \theta; x) &= f_{Y|X}(y | x_i; \theta) f(x, \theta) \\ &= c f(y; \theta) [F(y; \theta) - F(x_i; \theta)]^{s-1} \\ &\quad \times [1 - F(y; \theta)]^{R_i-s} \prod_{j=1}^m f(x_j; \theta) \\ &\quad \times \prod_{\substack{j=1 \\ j \neq i}}^m [1 - F(x_j; \theta)]^{R_j}, \quad y \geq x_i, \end{aligned} \quad (10)$$

where  $c$  denotes a constant factor. Apart from a constant term, the predictive log-likelihood function is

$$\begin{aligned}
\ln L(y, \theta; x) &= \ln f(y, \theta) + (s-1) \ln [F(y; \theta) - F(x_i; \theta)] \\
&\quad + (R_i - s) \ln [1 - F(y; \theta)] + \sum_{j=1}^m \ln f(x_j; \theta) \\
&\quad + \sum_{\substack{j=1 \\ j \neq i}}^m R_j \ln [1 - F(x_j; \theta)], \quad y \geq x_i.
\end{aligned} \tag{11}$$

From Equations (1), (5) and (11), the log PLF of  $Y = y$  and  $\theta$ , for  $y \geq x_i$ , is given by

$$\begin{aligned}
\ln L(y, \theta; x) &= (m+1) \ln \theta + \ln \left[ \frac{f_0(y)}{\bar{F}_0(y)} \right] \\
&\quad + (s-1) \ln \left[ 1 - \left( \frac{\bar{F}_0(y)}{\bar{F}_0(x_i)} \right)^\theta \right] \\
&\quad + \theta(R_i - s + 1) [\ln \bar{F}_0(y) - \ln \bar{F}_0(x_i)] \\
&\quad + \theta \sum_{j=1}^m (R_j + 1) \ln \bar{F}_0(x_j).
\end{aligned} \tag{12}$$

By using (12), the predictive likelihood equations (PLEs) for  $y$  and  $\theta$  (for  $y \geq x_i$ ) are given by

$$\begin{aligned}
\frac{\partial}{\partial y} \ln L_p(y, \theta) &= -\frac{1}{f(x_{(m)})} [f(y) \{f_0(y) + f_\theta(y)\} - \theta(R_{i+1} + 1)f_0(y)] \\
&\quad + \theta(s-1)f_\theta(y) \left[ \frac{f_0(y)}{1 - \frac{f_\theta(y)}{f_0(y)}} \right] \\
&= 0,
\end{aligned} \tag{13}$$

and

$$\begin{aligned}
\frac{\partial}{\partial \theta} \ln L_p(y, \theta) &= \frac{m+1}{\theta} - (s-1) \left[ \frac{f_\theta(y)}{f_0(y)} \right] \\
&\quad + (R_{i+1} + 1) \ln[f_0(y)] - \ln[f_\theta(y)] \\
&\quad + \sum_{i=1}^m (R_i + 1) \ln[f_0(x_i)] \\
&= 0.
\end{aligned} \tag{14}$$

**Example 1. (i) (Exponential distribution):** For the case of exponential distribution, we have

$$f_0(x) = e^{-x}, \quad x > 0. \tag{15}$$

In this case, the PLEs reduce to

$$\frac{\partial}{\partial y} \ln L_p(y, \theta) = -\theta(R_{i+1} + 1) + (\theta s - 1)e^{-\theta(y-x_i)} = 0, \tag{16}$$

and

$$\begin{aligned}
\frac{\partial}{\partial \theta} \ln L_p(y, \theta) &= \frac{m+1}{\theta} - \left[ \sum_{i=1}^m (R_i + 1)x_i + (R_{i+1} + 1)(y - x_i) \right] \\
&\quad + (s-1) \frac{y - x_i}{1 - e^{-\theta(y-x_i)}} \\
&= 0.
\end{aligned} \tag{17}$$

Now, the MLP of  $y$  and PMLE of  $\theta$  can be obtained as

$$\hat{y}_{MLP} = x_i + \frac{1}{\theta} \ln \left[ \frac{R_{i+1}}{R_{i+1} + 1} \right], \tag{18}$$

and

$$\hat{\theta}_{PMLE} = \frac{m+1}{T_0(x)}. \tag{19}$$

**Example (ii) (Lindley distribution):** For the case of the Lindley distribution, we have

$$f_0(x) = \frac{1}{2}(1+x)e^{-x}, \quad x > 0. \tag{20}$$



## Log-Likelihood

$$\begin{aligned}\ln L(y, x, \theta, \lambda) = & (m+1) \ln \theta + \ln \left\{ \frac{\lambda^2}{\lambda+1+\lambda y} (1+y)^2 \right\} \\ & + (s-1) \ln \left[ 1 - \left( \frac{\lambda+1+\lambda y}{\lambda+1+\lambda x_i} \right)^\theta e^{-\lambda \theta (y-x_i)} \right] \\ & + \theta (R_i - s + 1) [\ln(\lambda+1+\lambda y) - \lambda y - \ln(\lambda+1+\lambda x_i) + \lambda x_i]\end{aligned}$$

Derivative w.r.t.  $y$ :

$$\begin{aligned}\frac{\partial L(y, x, \theta, \lambda)}{\partial y} &= 0 \\ \Rightarrow \frac{1 - \lambda(y-1)(\lambda+1+\lambda y) + \frac{\lambda^2}{\lambda+1}(1+y^2)}{(1+y)(\lambda+1+\lambda y)} - \frac{(R_i - s - 1)\lambda^3(1+y)}{\lambda+1+\lambda y} \\ &+ \frac{\theta(s-1)\lambda^2(1+y) \left( \frac{\lambda+1+\lambda y}{\lambda+1+\lambda x_i} \right)^{\theta-1} e^{-\lambda \theta (y-x_i)}}{1 - \left\{ \left( \frac{\lambda+1+\lambda y}{\lambda+1+\lambda x_i} \right)^\theta e^{-\lambda \theta (y-x_i)} \right\}} = 0\end{aligned}$$

Derivative w.r.t.  $\theta$ :

$$\begin{aligned}\frac{\partial \ln L(y, \theta)}{\partial \theta} &= 0 \\ \Rightarrow \frac{m+1}{\theta} - (s-1) \ln \left[ \frac{F_0(y)}{F_0(x_i)} \right] &\left( \frac{\left[ \frac{F_0(y)}{F_0(x_i)} \right]^\theta}{1 - \left[ \frac{F_0(y)}{F_0(x_i)} \right]^\theta} \right) + (R_i - s + 1) [\ln(F_0(y)) - \ln(F_0(x_i))] \\ &+ \sum_{j=1}^m (R_j + 1) \ln[F_0(x_j)] = 0\end{aligned}$$

$$\begin{aligned}
&\Rightarrow \frac{m+1}{\theta} - (s-1) \ln \left\{ \left( \frac{\lambda+1+\lambda y}{\lambda+1+\lambda x_i} \right) e^{-\lambda(y-x_i)} \right\} \times \frac{\left[ \left( \frac{\lambda+1+\lambda y}{\lambda+1+\lambda x_i} \right)^\theta e^{-\lambda\theta(y-x_i)} \right]}{1 - \left[ \left( \frac{\lambda+1+\lambda y}{\lambda+1+\lambda x_i} \right)^\theta e^{-\lambda\theta(y-x_i)} \right]} \\
&+ (R_i - s + 1) \left[ \ln \left\{ \left( \frac{\lambda+1+\lambda y}{\lambda+1} \right) e^{-\lambda y} \right\} - \ln \left\{ \left( \frac{\lambda+1+\lambda x_i}{\lambda+1} \right) e^{-\lambda x_i} \right\} \right] \\
&+ \sum_{j=1}^m (R_j + 1) \ln \left\{ \left( \frac{\lambda+1+\lambda x_j}{\lambda+1} \right) e^{-\lambda x_j} \right\} = 0
\end{aligned}$$

**Derivative w.r.t.  $\lambda$**

$$\frac{\partial \ln L(y, x, \theta, \lambda)}{\partial \lambda} = 0$$

**Example (iii) (Rayleigh distribution):** For the case of Rayleigh distribution, we have

$$f_0(x) = 2xe^{-x^2}, \quad x > 0. \quad (21)$$

For the case of the Rayleigh distribution, we have

$$\begin{aligned}
\ln L(y, \theta, \lambda) &= (m+1) \ln \theta + \ln(2\lambda y) + (s-1) \ln \left[ 1 - e^{-\lambda\theta(x_i^2+y^2)} \right] \\
&+ \theta(R_i - s + 1) [-\lambda y^2 + \lambda x_i^2] - \lambda\theta \sum_{j=1}^m (R_j + 1) x_j^2
\end{aligned}$$

**Derivative with respect to  $\theta$ :**

$$\frac{\partial \ln L(y, \theta, \lambda)}{\partial \theta} = 0$$

$$\begin{aligned} &\Rightarrow \frac{m+1}{\theta} - (s-1)\lambda(y^2 + x_i^2) \frac{e^{-\lambda\theta(y^2+x_i^2)}}{1 - e^{-\lambda\theta(y^2+x_i^2)}} \\ &+ (R_i - s + 1)\lambda(x_i^2 - y^2) + \sum_{j=1}^m (R_j + 1)(-\lambda x_j^2) = 0 \end{aligned}$$

**Derivative with respect to  $y$ :**

$$\begin{aligned} &\frac{\partial \ln L(y, \theta, \lambda)}{\partial y} = 0 \\ &\Rightarrow \frac{1}{y} + (-2\lambda y) + (R_i - s + 1)\theta(2\lambda y) \\ &+ (s-1)2\lambda y \frac{e^{-\lambda\theta(y^2+x_i^2)}}{1 - e^{-\lambda\theta(y^2+x_i^2)}} = 0 \end{aligned}$$

**Derivative with respect to  $\lambda$ :**

$$\begin{aligned} &\frac{\partial \ln L(y, \theta, \lambda)}{\partial \lambda} = 0 \\ &\Rightarrow \frac{1}{\lambda} + (s-1)\theta(x_i^2 + y^2) \frac{e^{-\lambda\theta(x_i^2+y^2)}}{1 - e^{-\lambda\theta(x_i^2+y^2)}} \\ &+ \theta(R_i - s + 1)(-y^2 + x_i^2) - \theta \sum_{j=1}^m (R_j + 1)x_j^2 = 0 \end{aligned}$$

## 2.2 Best Unbiased Predictor

A predictor  $\hat{Y}$  for the unobserved quantity  $Y = X_{i,(s)}$  is said to be a best unbiased predictor (BUP) if it satisfies two important criteria. First, the prediction error must be unbiased, that is,

$$E(\hat{Y} - Y) = 0.$$

Second, among all predictors satisfying the unbiasedness condition,  $\hat{Y}$  must yield the smallest prediction error variance. In other words,

$$\text{Var}(\hat{Y} - Y)$$

should not exceed the corresponding variance associated with any other unbiased predictor of  $Y$ . Thus, the BUP is the unbiased predictor that minimises the uncertainty in predicting the unobserved lifetime  $Y$ .

Since the conditional distribution of  $Y$  given  $\mathbf{X} = (X_1, \dots, x_i)$  is just the distribution of  $Y$  given  $X_i$ , therefore the BUP of  $Y$  is (see Nayak (2000))

$$\hat{Y}_{\text{BUP}} = E(Y \mid X_i = x_i).$$

By (9), we have

$$\begin{aligned} \hat{Y}_{\text{BUP}} &= \int_{x_i}^{\infty} y f_{Y|\theta}(y \mid x) dy \\ &= \int_{x_i}^{\infty} y \frac{\theta^{R_{i+1}+1} (y - x_i)^{R_{i+1}}}{\text{Beta}(R_{i+1} + 1, s)} dy. \end{aligned} \quad (22)$$

If the parameter  $\theta$  is unknown, we can replace  $\theta$  by its MLE and obtain an approximate BUP of  $Y$ .

**Example 2. (i) (Exponential distribution):** For exponential distribution with  $f_0(x) = e^{-x}$  and after replacing  $\theta$  by its MLE, we obtain the approximate BUP of  $Y$  as

$$\begin{aligned} \hat{Y}_{\text{BUP}} &= \int_{x_i}^{\infty} y \frac{\hat{\theta}^{R_{i+1}+1} (y - x_i)^{R_{i+1}}}{\text{Beta}(R_{i+1} + 1, s)} e^{-\hat{\theta}(y-x_i)} dy \\ &= x_i + \frac{1}{\hat{\theta}} \cdot \frac{E(U)}{E(1)} \\ &= x_i + \frac{1}{\hat{\theta}} E(U) \\ &= x_i + \frac{T_0(x)}{m+1} E(U). \end{aligned} \quad (23)$$

where  $U$  has the  $\text{Beta}(R_{i+1} + 1, s)$  distribution and  $Z_m$  denotes the  $m$ -th order statistic out of  $R_i$  units from a standard exponential distribution.

**(ii) (Lindley distribution):**

$$F_0^{-1}(y) = -\frac{1}{\theta} W \{ -y(\theta + 1)e^{-(\theta+1)} \} - (\theta + 1),$$

where  $W(\cdot)$  is the Lambert W function and

$$\begin{aligned}
\hat{Y}_{\text{BUP}} &= \int_0^1 F_0^{-1}\left(u^{\frac{1}{\theta_{\text{MLE}}}} \bar{F}_0(x_i)\right) \frac{u^{R_i-s}(1-u)^{s-1}}{\text{Beta}(R_i-s+1, s)} du, \\
&= \int_0^1 \left[ -\frac{1}{\theta} W\left\{ -u^{\frac{1}{\theta_{\text{MLE}}}} \left( \frac{\theta+1+\theta x_i}{\theta+1} \right) e^{-\theta x_i} (\theta+1) e^{-(\theta+1)} \right\} - (\theta+1) \right] \\
&\quad \times \frac{u^{R_i-s}(1-u)^{s-1}}{\text{Beta}(R_i-s+1, s)} du \tag{24}
\end{aligned}$$

**(iii) (Rayleigh distribution):** For the Rayleigh distribution

$$\begin{aligned}
\bar{F}_0(x) &= e^{-\lambda x^2} \\
y &= e^{-\lambda x^2}
\end{aligned}$$

Taking logarithm:

$$\begin{aligned}
\ln y &= -\lambda x^2 \\
x^2 &= -\frac{1}{\lambda} \ln y \\
x &= \sqrt{-\frac{1}{\lambda} \ln y}
\end{aligned}$$

## Inverse Transformation

$$F_0^{-1}(y) = \sqrt{-\frac{1}{\lambda} \ln y}$$

$$\begin{aligned}
\hat{Y}_{\text{BHP}} &= \int_0^1 \sqrt{\left(-\frac{1}{\lambda}\right) \ln\left(u^{\frac{1}{\theta_{\text{MLE}}}} e^{-\lambda x_i^2}\right)} \frac{u^{R_i-s}(1-u)^{s-1}}{\text{Beta}(R_i-s+1, s)} du \\
&= \int_0^1 \sqrt{\left(-\frac{1}{\lambda}\right) (-\lambda x_i^2) + \frac{1}{\theta_{\text{MLE}}} \left(-\frac{1}{\lambda}\right) \ln u} \frac{u^{R_i-s}(1-u)^{s-1}}{\text{Beta}(R_i-s+1, s)} du \\
&= \int_0^1 \sqrt{x_i^2 - \frac{1}{\lambda \theta_{\text{MLE}}} \ln u} \frac{u^{R_i-s}(1-u)^{s-1}}{\text{Beta}(R_i-s+1, s)} du \tag{25}
\end{aligned}$$

## 2.3 Conditional Median Predictor (CMP)

The Conditional Median Predictor (CMP) of  $Y$  is defined as the median of the conditional distribution of  $Y$  given  $X_i = x_i$ ; that is,

$$P_\theta(Y \leq \hat{Y} \mid X_i = x_i) = P_\theta(Y \geq \hat{Y} \mid X_i = x_i).$$

Using the relation

$$P_\theta(Y \leq \hat{Y} \mid X_i = x_i) = P_\theta \left[ \left( \frac{\bar{F}_0(Y)}{\bar{F}_0(X_i)} \right)^\theta \geq \left( \frac{\bar{F}_0(\hat{Y})}{\bar{F}_0(X_i)} \right)^\theta \mid X_i = x_i \right],$$

and using the fact that the distribution of  $\left( \frac{\bar{F}_0(Y)}{\bar{F}_0(X_i)} \right)^\theta$  given  $X_i = x_i$  is a Beta( $R_i - s + 1, s$ ) distribution, we obtain the CMP of  $Y$  as

$$\hat{Y}_{\text{CMP}} = \bar{F}_0^{-1} \left( \bar{F}_0(x_i) [\text{Med}(U)]^{\frac{1}{s}} \right),$$

where  $U$  has Beta( $R_i - s + 1, s$ ) distribution and  $\text{Med}(U)$  stands for the median of  $U$ . By substituting  $\theta$  with its MLE, we obtain

$$\hat{Y}_{\text{CMP}} = \bar{F}_0^{-1} \left( \bar{F}_0(x_i) (\text{Med}(U))^{\frac{T_0(X)}{m}} \right),$$

as the CMP of  $Y$ .

### Example 3

(i) **Exponential distribution:** Taking  $\bar{F}_0(x) = e^{-x}$ , for the case of exponential distribution, we obtain the CMP of  $Y$  as

$$\begin{aligned} \hat{Y}_{\text{CMP}} &= -\ln \left( e^{-x} [\text{Med}(U)]^{\frac{T_0(X)}{m}} \right) \\ &= x_i - \frac{T_0(X)}{m} \ln[\text{Med}(U)] \\ &= x_i + \frac{T_0(X)}{m} \text{Med}(Z_{S, R_i}). \end{aligned} \tag{26}$$

(ii) **Lindley distribution:**

$$\bar{F}_0(x) = \frac{(\lambda + 1 + \lambda x)}{(\lambda + 1)} e^{-\lambda x}$$

$$\bar{F}_0^{-1}(x) = -\frac{1}{\lambda} W \{ -y(\lambda + 1)e^{-(\lambda+1)} \} - (\lambda + 1)$$

$$\hat{Y}_{\text{CMP}} = -\frac{1}{\lambda} W \left[ -\frac{(\lambda + 1 + \lambda x_i)}{(\lambda + 1)} e^{-\lambda x_i} (\lambda + 1) e^{-(\lambda+1)} \{ \text{med}(u) \}^{\frac{1}{\hat{\theta}_{\text{MLE}}}} \right] - (\lambda + 1)$$

For  $\lambda = 1$ ,

$$\begin{aligned} \hat{Y}_{\text{CMP}} &= -W \left[ -\frac{(2 + x_i)}{2} e^{-x_i} \cdot 2e^{-2} \{ \text{med}(u) \}^{\frac{1}{\hat{\theta}_{\text{MLE}}}} \right] - 2 \\ &= -W \left[ -(2 + x_i) e^{-x_i} e^{-2} \{ \text{med}(u) \}^{\frac{1}{\hat{\theta}_{\text{MLE}}}} \right] - 2. \end{aligned} \quad (27)$$

**(iii) Rayleigh Distribution :**

$$\bar{F}_0(x) = e^{-\lambda x^2}$$

Let

$$y = e^{-\lambda x^2}$$

Then,

$$x = \sqrt{-\frac{1}{\lambda} \ln y}$$

$$\hat{Y}_{\text{CMP}} = \bar{F}_0^{-1} \left[ \bar{F}_0(x_i) (\text{Med}(U))^{\frac{T(x)}{m}} \right]$$

Substituting  $\bar{F}_0(x) = e^{-\lambda x^2}$ :

$$\hat{Y}_{\text{CMP}} = \sqrt{-\frac{1}{\lambda} \ln \left( e^{-\lambda x_i^2} \cdot (\text{Med}(U))^{\frac{T(x)}{m}} \right)}$$

Using log properties:

$$\hat{Y}_{\text{CMP}} = \sqrt{-\frac{1}{\lambda} \left[ -\lambda x_i^2 + \frac{T(x)}{m} \ln(\text{Med}(U)) \right]}$$

Let  $\lambda = 1$

$$\hat{Y}_{\text{CMP}} = \sqrt{x_i^2 - \frac{T(x)}{m\lambda} \ln(\text{Med}(U))} \quad (28)$$

## 2.4 Bayesian Predictors

In this section, Bayesian methodology is employed to predict the unobserved lifetime

$$Y = X_{i,(s)}, \quad s = 1, 2, \dots, R_i, \quad i = 1, 2, \dots, m,$$

using the information contained in the observed progressively Type-II right censored sample

$$\mathbf{x} = (x_1, x_2, \dots, x_m).$$

Bayesian prediction relies on the predictive distribution of  $Y$  conditional on the observed data, denoted by  $f^*(y|x)$ , together with a specified loss function. Let  $\hat{Y} = \delta(X)$  be a predictor of  $Y$ . If  $y$  is the realised value of the quantity of interest and  $\hat{y}$  is its prediction, then the discrepancy associated with the prediction is quantified through a loss function  $L(y, \hat{y})$ .

A commonly adopted criterion is the squared error loss (SEL), defined by

$$L(y, \hat{y}) = (\hat{y} - y)^2.$$

Because of its symmetric nature and analytical convenience, this loss function is frequently used in Bayesian inference. Under SEL, the Bayes predictor is the posterior predictive mean, namely

$$\hat{y}_{SEP} = E(Y|x).$$

Another popular symmetric loss function is the absolute difference loss (ADL),

$$L(y, \hat{y}) = |y - \hat{y}|.$$

For this loss structure, the Bayes predictor is given by the median of the predictive distribution of  $Y$ , which will be denoted by  $\hat{y}_{ADP}$ .

In reliability analysis and life-testing studies, overestimation and underestimation often incur unequal costs. Consequently, symmetric loss functions may not adequately reflect practical considerations. To address this issue, an asymmetric loss function known as the general entropy loss (GEL) is often employed:

$$L(y, \hat{y}) \propto \left(\frac{\hat{y}}{y}\right)^q - q \ln\left(\frac{\hat{y}}{y}\right) - 1, \quad q \neq 0. \quad (29)$$

The corresponding Bayes predictor under the GEL criterion is

$$\hat{y}_{GEP} = [E(Y^{-q} | x)]^{-1/q}. \quad (30)$$



To proceed with Bayesian inference, suppose that the unknown parameter  $\theta$  is assigned a Gamma prior distribution with shape parameter  $\alpha$  and rate parameter  $\beta$ . Its prior density is given by

$$\pi(\theta) = \frac{\beta^\alpha}{\Gamma(\alpha)} \theta^{\alpha-1} e^{-\beta\theta}, \quad \theta > 0. \quad (31)$$

Combining this prior information with the likelihood function yields the posterior density of  $\theta$  as

$$\pi(\theta|x) = \frac{[\beta + T(x)]^{m+\alpha}}{\Gamma(m+\alpha)} \theta^{m+\alpha-1} e^{-\theta[\beta+T(x)]}, \quad (32)$$

which is again a Gamma distribution with updated parameters determined by the observed data.

The Bayes predictive density function of  $Y$  given  $X_i = x_i$  is

$$f^*(y|x_i) = \int_0^\infty f(y|x_i, \theta) \pi(\theta|x) d\theta. \quad (33)$$

Substituting, we obtain

$$\begin{aligned} f^*(y|x_i) &= \int_0^\infty s \binom{R_i}{s} \theta \frac{f_0(y)}{\bar{F}_0(y)} [\bar{F}_0(y)]^{\theta(R_i-s+1)} \\ &\quad \times \left[ (\bar{F}_0(x_i))^\theta - (\bar{F}_0(y))^\theta \right]^{s-1} [\bar{F}_0(x_i)]^{-\theta R_i} \\ &\quad \times \frac{[\beta + T(x)]^{m+\alpha}}{\Gamma(m+\alpha)} \theta^{m+\alpha-1} e^{-\theta[T(x)+\beta]} d\theta, \\ &\quad y \geq x_i. \end{aligned} \quad (34)$$

Using binomial expansion,

$$\left[ (\bar{F}_0(x_i))^\theta - (\bar{F}_0(y))^\theta \right]^{s-1} = \sum_{j=0}^{s-1} \binom{s-1}{j} (-1)^j [\bar{F}_0(y)]^{\theta j} [\bar{F}_0(x_i)]^{\theta(s-j-1)}. \quad (35)$$

Thus,

$$\begin{aligned} f^*(y|x_i) &= s \binom{R_i}{s} \frac{f_0(y)}{\bar{F}_0(y)} \frac{m+\alpha}{\beta + T(x)} \sum_{j=0}^{s-1} \binom{s-1}{j} (-1)^j \\ &\quad \times \left[ 1 - \frac{(R_i - s + j + 1) \ln \left( \frac{\bar{F}_0(y)}{\bar{F}_0(x_i)} \right)}{\beta + T(x)} \right]^{-(m+\alpha+1)}. \end{aligned} \quad (36)$$

Under ADL, the Bayes predictor  $\hat{Y}_{ADP}$  satisfies

$$\begin{aligned}
\frac{1}{2} &= \int_{x_i}^{\hat{Y}_{ADP}} f^*(y|x_i) dy \\
&= s \binom{R_i}{s} \sum_{j=0}^{s-1} \binom{s-1}{j} (-1)^j \frac{1}{R_i - s + j + 1} \\
&\quad \times \left[ 1 - \left( 1 - \frac{(R_i - s + j + 1) \ln \left( \frac{\bar{F}_0(\hat{Y}_{ADP})}{\bar{F}_0(x_i)} \right)}{\beta + T(x)} \right) \right]^{-(m+\alpha)}. \quad (37)
\end{aligned}$$

Using (30), the Bayes point predictor of  $Y$  under GEL loss function is given by

$$\begin{aligned}
\hat{Y}_{GEP} &= \left( \int_{x_i}^{\infty} y^{-q} f^*(y|x_i) dy \right)^{-1/q} \\
&= \left( s \binom{R_i}{s} \frac{m + \alpha}{\beta + T(x)} \sum_{j=0}^{s-1} \binom{s-1}{j} (-1)^j I(j) \right)^{-1/q}. \quad (38)
\end{aligned}$$

where

$$I(j) = \int_{x_i}^{\infty} y^{-q} \frac{f_0(y)}{\bar{F}_0(y)} \left[ 1 - \frac{(R_i - s + j + 1) \ln \left( \frac{\bar{F}_0(y)}{\bar{F}_0(x_i)} \right)}{\beta + T(x)} \right]^{-(m+\alpha+1)} dy.$$

Note that the Bayes point predictor  $\hat{Y}_{SEP}$  under SEL can be obtained by setting  $q = -1$  in (38).

For a special case, when  $s = 1$ , the equations (37) and (38) are reduced to:

$$\frac{1}{2} = \left[ 1 - \frac{R_i}{\beta + T(x)} \ln \left( \frac{\bar{F}_0(\hat{Y}_{ADP})}{\bar{F}_0(x_i)} \right) \right]^{-(m+\alpha)}$$

and

$$\hat{Y}_{GEP} = \left( \frac{m + \alpha}{\beta + T(x)} R_i I(0) \right)^{-1/q},$$

where

$$I(0) = \int_{x_i}^{\infty} y^{-q} \frac{f_0(y)}{\bar{F}_0(y)} \left[ 1 - \frac{R_i \ln \left( \frac{\bar{F}_0(y)}{\bar{F}_0(x_i)} \right)}{\beta + T(x)} \right]^{-(m+\alpha+1)} dy.$$

In this case, we obtain the Bayes point predictor  $\hat{Y}_{ADP}$  as

$$\hat{Y}_{ADP} = \bar{F}_0^{-1} \left( \bar{F}_0(x_i) \exp \left[ \frac{\beta + T(x)}{m + \alpha} \left( 1 - \left( \frac{1}{2} \right)^{\frac{1}{m+\alpha}} \right) \right] \right).$$

**Example 4.** (i) (Exponential distribution): For the exponential distribution with  $\bar{F}_0(x) = e^{-x}$ , the Bayes point predictors  $\hat{Y}_{ADP}$  and  $\hat{Y}_{GEP}$  of  $Y$  can be obtained from the following equations:

$$\begin{aligned} \frac{1}{2} &= s \binom{R_i}{s} \sum_{j=0}^{s-1} \binom{s-1}{j} (-1)^j \frac{1}{R_i - s + j + 1} \\ &\times \left[ 1 - \left( 1 + \frac{(R_i - s + j + 1)(\hat{Y}_{ADP} - x_i)}{\beta + T_0(x)} \right)^{-(m+\alpha)} \right], \end{aligned} \quad (39)$$

and

$$\hat{Y}_{GEP} = \left[ s \binom{R_i}{s} \frac{m + \alpha}{\beta + T_0(x)} \sum_{j=0}^{s-1} \binom{s-1}{j} (-1)^j I_0(j) \right]^{-\frac{1}{q}}, \quad (40)$$

where

$$I_0(j) = \int_{x_i}^{\infty} y^{-q} \left[ 1 + \frac{(R_i - s + j + 1)(y - x_i)}{\beta + T_0(x)} \right]^{-(m+\alpha+1)} dy.$$

(ii) Lindley Distribution:

$$\bar{F}_0(x) = \left( \frac{1 + \lambda + \lambda x}{1 + \lambda} \right) e^{-\lambda x}$$

For  $\lambda = 1$ ,

$$\bar{F}_0(x) = \frac{2 + x}{2} e^{-x}$$

$\hat{Y}_{ADP}$  can be obtained from the following equation:

$$\frac{1}{2} = s \binom{R_i}{s} \sum_{j=0}^{s-1} \binom{s-1}{j} (-1)^j \frac{1}{R_i - s + j + 1}$$

$$\times \left[ 1 - \left( 1 - \frac{(R_i - s + j + 1) \ln \left( \frac{\bar{F}_0(\hat{Y}_{ADP})}{\bar{F}_0(x_i)} \right)}{\beta + T(x)} \right)^{-(m+\alpha)} \right]$$

Substituting  $\bar{F}_0(x)$ :

$$= s \binom{R_i}{s} \sum_{j=0}^{s-1} \binom{s-1}{j} (-1)^j \frac{1}{R_i - s + j + 1} \\ \times \left[ 1 - \left\{ 1 - \frac{(R_i - s + j + 1) \ln \left( \frac{\frac{2+\hat{Y}_{ADP}}{2} e^{-\hat{Y}_{ADP}}}{\frac{2+x_i}{2} e^{-x_i}} \right)}{\beta + T(x)} \right\}^{-(m+\alpha)} \right]$$

Simplifying:

$$= s \binom{R_i}{s} \sum_{j=0}^{s-1} \binom{s-1}{j} (-1)^j \frac{1}{R_i - s + j + 1} \\ \times \left[ 1 - \left\{ 1 - \frac{(R_i - s + j + 1) \left[ \ln(2 + \hat{Y}_{ADP}) - \hat{Y}_{ADP} - \ln(2 + x_i) + x_i \right]}{\beta + T(x)} \right\}^{-(m+\alpha)} \right] \\ \hat{Y}_{GEP} = \left[ s \binom{R_i}{s} \frac{m + \alpha}{\beta + T(x)} \sum_{j=0}^{s-1} \binom{s-1}{j} (-1)^j I(j) \right]^{\frac{1}{2}}$$

$$I(j) = \int_{x_i}^{\infty} y^{-2} \frac{f_0(y)}{\bar{F}_0(y)} \left[ 1 - \left( \frac{(R_i - s + j + 1) \ln \left( \frac{\bar{F}_0(y)}{\bar{F}_0(x_i)} \right)}{\beta + T(x)} \right) \right]^{-(m+\alpha+1)} dy$$

Substituting

$$f_0(y) = \frac{1}{2}(1+y)e^{-y}, \quad \bar{F}_0(y) = \frac{2+y}{2}e^{-y},$$

we get

$$I(j) = \int_{x_i}^{\infty} y^{-2} \frac{\frac{1}{2}(1+y)e^{-y}}{\frac{2+y}{2}e^{-y}} \left[ 1 - \frac{(R_i - s + j + 1)}{\beta + T(x)} \ln \left( \frac{\frac{2+y}{2}e^{-y}}{\frac{2+x_i}{2}e^{-x_i}} \right) \right]^{-(m+\alpha+1)} dy$$

Simplifying,

$$I(j) = \int_{x_i}^{\infty} y^{-2} \left( \frac{1+y}{2+y} \right) \left[ 1 - \frac{(R_i - s + j + 1)}{\beta + T(x)} \{ \ln(2+y) - y - \ln(2+x_i) + x_i \} \right]^{-(m+\alpha+1)} dy$$

Or equivalently,

$$I(j) = \int_{x_i}^{\infty} y^{-2} \left( \frac{1+y}{2+y} \right) \left[ 1 - \frac{(R_i - s + j + 1) \{ \ln(2+y) - (y - x_i) - \ln(2+x_i) \}}{\beta + T(x)} \right]^{-(m+\alpha+1)} dy$$

(iii) Rayleigh Distribution:

$$\begin{aligned} \bar{F}_0(x) &= e^{-\lambda x^2}, \\ f_0(x) &= 2\lambda x e^{-\lambda x^2} \end{aligned}$$

For  $\lambda = 1$ :

$$\begin{aligned} \bar{F}_0(x) &= e^{-x^2}, \\ f_0(x) &= 2x e^{-x^2} \end{aligned}$$

**ADP can be obtained from the following equation:**

$$\begin{aligned} \frac{1}{2} &= s \binom{R_i}{s} \sum_{j=0}^{s-1} \binom{s-1}{j} (-1)^j \frac{1}{R_i - s + j + 1} \\ &\times \left[ 1 - \left\{ 1 + \frac{(R_i - s + j + 1) \ln \left( \frac{\bar{F}_0(Y_{ADP})}{\bar{F}_0(x_i)} \right)}{\beta + T(x)} \right\}^{-(m+\alpha)} \right] \end{aligned} \quad (41)$$

Substituting  $\bar{F}_0(x) = e^{-x^2}$ :

$$\begin{aligned} \frac{1}{2} &= s \binom{R_i}{s} \sum_{j=0}^{s-1} \binom{s-1}{j} (-1)^j \frac{1}{R_i - s + j + 1} \\ &\times \left[ 1 - \left\{ 1 + \frac{(R_i - s + j + 1) \ln \left( \frac{e^{-Y_{ADP}^2}}{e^{-x_i^2}} \right)}{\beta + T(x)} \right\}^{-(m+\alpha)} \right] \end{aligned}$$

Simplifying the logarithm:

$$\begin{aligned} \frac{1}{2} &= s \binom{R_i}{s} \sum_{j=0}^{s-1} \binom{s-1}{j} (-1)^j \frac{1}{R_i - s + j + 1} \\ &\times \left[ 1 - \left\{ 1 + \frac{(R_i - s + j + 1)(-Y_{ADP}^2 + x_i^2)}{\beta + T(x)} \right\}^{-(m+\alpha)} \right] \\ \hat{Y}_{GEP} &= \left[ s \binom{R_i}{s} \frac{m + \alpha}{\beta + T(x)} \sum_{j=0}^{s-1} \binom{s-1}{j} (-1)^j I(j) \right]^{\frac{1}{\alpha}} \end{aligned} \quad (42)$$

where,

$$\begin{aligned} I(j) &= \int_{x_i}^{\infty} y^{-\alpha} \frac{f_0(y)}{\bar{F}_0(y)} \left[ 1 - \frac{(R_i - s + j + 1) \ln \left( \frac{\bar{F}_0(y)}{\bar{F}_0(x_i)} \right)}{\beta + T(x)} \right]^{-(m+\alpha+1)} dy \\ \text{i.e., } I(j) &= \int_{x_i}^{\infty} y^{-\alpha} \frac{2ye^{-y^2}}{e^{-y^2}} \left[ 1 - \frac{(R_i - s + j + 1) \ln \left( \frac{e^{-y^2}}{e^{-x_i^2}} \right)}{\beta + T(x)} \right]^{-(m+\alpha+1)} dy \\ \text{i.e., } I(j) &= \int_{x_i}^{\infty} y^{-\alpha} \cdot 2y \left[ 1 - \frac{(R_i - s + j + 1)(x_i^2 - y^2)}{\beta + T(x)} \right]^{-(m+\alpha+1)} dy \\ \text{i.e., } I(j) &= \int_{x_i}^{\infty} 2y^{1-\alpha} \left[ 1 - \frac{(R_i - s + j + 1)(x_i^2 - y^2)}{\beta + T(x)} \right]^{-(m+\alpha+1)} dy \end{aligned}$$

### 3 Simulation Study and Discussion

The expressions of the predictors indicate that an analytic comparison is not possible. Therefore, a Monte Carlo simulation study is used to evaluate and compare the performance with respect to their biases and mean-squared prediction errors (MSPEs) for the predictors MLP, BUP, CMP, ADP, SEP, and BGP. We randomly generated 1000 progressively censored samples from exponential, Lindley and Rayleigh distributions with  $\theta = 0.75, 1, 2$ . We also used the censoring scheme  $R_1 = (0, 0, 3, 0, 3, 0, 0, 5)$ . To compute Bayesian predictors, we assume a gamma prior with  $\alpha = 2$  and  $\beta = 1$ . Tables 5 – 13 display the biases and MSPEs of different predictors obtained from this simulation study. All the computations are performed using the R software package.

We consider

$$F(x, \theta) = 1 - \bar{F}_0^\theta(x), \quad x > 0, \theta > 0, \quad (43)$$

with

1.  $\bar{F}_0(x) = e^{-x}$ , the standard exponential distribution,
2.  $\bar{F}_0(x) = (1 + \frac{x}{2})e^{-x}$ , the standard Lindley distribution,
3.  $\bar{F}_0(x) = e^{-x^2}$ , the standard Rayleigh distribution.

Tables 5 – 7 display the biases and MSPEs for the said predictors for the exponential base distribution for the specified values of  $\theta$ . For a comprehensive visual display, we have placed the pictorial views in Figures 3-5. It has been observed that the BUPs produce the best results in terms of biases and MSPEs across the censoring scheme and all values of  $\theta$  considered. The SEPs are the second-best predictors. We also note that the MLP does not work well. However, except for BUPs and SEPs, all other predictors are underestimated, and over time, the biases and MSPEs are decreasing, indicating higher efficiency at large failure points.

Tables 8 – 10 display the biases and MSPEs for the said predictors for the Lindley base distribution for the specified values of  $\theta$ . For a comprehensive visual display, we have placed the pictorial views in Figures 6-8. Overall, similar observations are noticed as those for the exponential model. However, the predictors are more efficient than those for exponential base distributions in the proportional hazards model, in terms of bias and MSPE.

Tables 11–13 display the biases and MSPEs for the said predictors for the Rayleigh base distribution for the specified values of  $\theta$ . For a comprehensive visual display, we have placed the pictorial views in Figures 9–11. It has been noted that the observations are in the same tune as those in the cases of exponential and Lindley base proportional hazard models.

We observe that most of the predictors usually underpredict the life-lengths  $Y = X_{i:(s)}$  ( $s = 1, 2, \dots, R_i$ ;  $i = 1, 2, \dots, m$ ), except the BUP and SEP, which overpredict sometimes.

We note that for  $q$  close to  $-1$ , the GEPs are close to the SEPs. These tables show that the Bayes predictors relative to the asymmetric loss function (GE) are sensitive to the value of the shape parameter  $q$ .

If the number of censored units at different failure points is the same, the efficiency of each predictor is more or less the same. As the number of censored units increases, the predictors are estimated more efficiently. For example, in the studied censoring scheme, at  $x_3$  and  $x_5$ , the units censored is 3, the biases and MSPEs are similar, whereas at  $x_8$ , the censored units is 5, the biases are more precise and MSPEs are lower than that at failure points  $x_3$  and  $x_5$ , indicating estimating predictors better. Though only one censoring scheme may not be sufficient for making a decision, the study of more censoring schemes is needed to support the assertion.

In general, the biases and MSPEs for the Rayleigh model are lower than those of the exponential and Lindley models. The ordering is roughly Rayleigh  $\succ$  Exponential  $\succ$  Lindley.

## 4 Data Analysis

In this section, we present the results of numerical experiments on the different point predictors investigated in the previous sections, using both simulated (synthetic) and real-life data.

### 4.1 Synthetic Data Analysis

According to the following steps, the different point predictors are obtained as described in Section 2.

- (i) For given values of  $\alpha = 2$  and  $\beta = 1$ , we sample  $\theta = 1.952$  from the prior pdf (31).



- (ii) Using the value of  $\theta = 1.952$  from step (i), we generate a progressively Type-II censored sample of size  $m = 8$  with the censoring scheme

$$R_1 = (0, 0, 3, 0, 3, 0, 0, 5) \quad (44)$$

from the exponential distribution according to the algorithm presented in Balakrishnan and Aggarwala (2000).

In case of the exponential base distribution, the sample generated is

0.0322 0.1201 0.1249 0.1435 0.1633 0.1805 0.2101 0.2158

In the case of the Lindley base distribution, the sample generated is

0.0119 0.4093 0.4351 0.6747 0.0813 0.4174 0.6201 0.9957

In the case of the Rayleigh base distribution, the sample generated is

0.0774 0.4724 0.4881 0.6197 0.2036 0.4774 0.5916 0.7692

- (iii) Using these samples, we obtain the non-Bayesian point predictors MLP, BUP and CMP and Bayesian point predictors ADP, SEP and GEP for  $X_{i:(s)}$  ( $s = 1, 2, \dots, R_i$ ;  $i = 3, 5, 8$ ). The results are displayed in Tables 1 – 3.

Table 1: Different Point Predictors for Exponential Distribution

|             | MLP   | BUP   | CMP   | ADP   | SEP   | GEP        |            |            |           |           |
|-------------|-------|-------|-------|-------|-------|------------|------------|------------|-----------|-----------|
|             |       |       |       |       |       | $q = -0.8$ | $q = -0.5$ | $q = -0.3$ | $q = 0.3$ | $q = 0.5$ |
| $X_{3:(1)}$ | 0.125 | 0.199 | 0.176 | 0.216 | 0.436 | 0.555      | 0.816      | 1.069      | 2.539     | 3.442     |
| $X_{3:(2)}$ | 0.209 | 0.311 | 0.280 | 0.400 | 0.785 | 0.885      | 1.088      | 1.268      | 2.156     | 2.632     |
| $X_{3:(3)}$ | 0.354 | 0.535 | 0.478 | 0.753 | 1.483 | 1.469      | 1.487      | 1.527      | 1.799     | 1.953     |
| $X_{5:(1)}$ | 0.163 | 0.238 | 0.215 | 0.256 | 0.499 | 0.621      | 0.879      | 1.120      | 2.413     | 3.155     |
| $X_{5:(2)}$ | 0.248 | 0.349 | 0.318 | 0.438 | 0.847 | 0.945      | 1.137      | 1.304      | 2.088     | 2.489     |
| $X_{5:(3)}$ | 0.393 | 0.573 | 0.516 | 0.791 | 1.546 | 1.521      | 1.524      | 1.551      | 1.765     | 1.889     |
| $X_{8:(1)}$ | 0.215 | 0.260 | 0.246 | 0.270 | 0.491 | 0.620      | 0.886      | 1.129      | 2.372     | 3.052     |
| $X_{8:(2)}$ | 0.262 | 0.316 | 0.300 | 0.362 | 0.666 | 0.788      | 1.026      | 1.231      | 2.185     | 2.660     |
| $X_{8:(3)}$ | 0.322 | 0.391 | 0.371 | 0.490 | 0.898 | 0.992      | 1.187      | 1.342      | 2.010     | 2.320     |
| $X_{8:(4)}$ | 0.322 | 0.391 | 0.371 | 0.490 | 0.898 | 0.992      | 1.187      | 1.342      | 2.010     | 2.320     |
| $X_{8:(5)}$ | 0.322 | 0.391 | 0.371 | 0.490 | 0.898 | 0.992      | 1.187      | 1.342      | 2.010     | 2.320     |

Table 2: Different Point Predictors for Lindley Distribution

|        | MLP     | BUP    | CMP    | ADP    | SEP    | GEP        |            |            |           |           |
|--------|---------|--------|--------|--------|--------|------------|------------|------------|-----------|-----------|
|        |         |        |        |        |        | $q = -0.8$ | $q = -0.5$ | $q = -0.3$ | $q = 0.3$ | $q = 0.5$ |
| X3,(1) | 0.43507 | 0.8590 | 0.7289 | 0.7024 | 0.8489 | 0.8298     | 0.8036     | 0.7876     | 0.7459    | 0.7338    |
| X3,(2) | 0.89341 | 1.4948 | 1.3165 | 1.2369 | 1.4696 | 1.4281     | 1.3695     | 1.3328     | 1.2336    | 1.2038    |
| X3,(3) | 1.67693 | 2.7665 | 2.4423 | 2.2658 | 2.7083 | 2.6190     | 2.4915     | 2.4106     | 2.1908    | 2.1230    |
| X5,(1) | 0.08131 | 0.5052 | 0.3751 | 0.3486 | 0.4951 | 0.4635     | 0.4194     | 0.3924     | 0.3236    | 0.3045    |
| X5,(2) | 0.53964 | 1.1410 | 0.9628 | 0.8832 | 1.1159 | 1.0609     | 0.9816     | 0.9308     | 0.7897    | 0.7465    |
| X5,(3) | 1.32317 | 2.4127 | 2.0886 | 1.9120 | 2.3549 | 2.2514     | 2.1016     | 2.0052     | 1.7349    | 1.6498    |
| X8,(1) | 0.99574 | 1.2501 | 1.1720 | 1.1561 | 1.2440 | 1.2388     | 1.2314     | 1.2267     | 1.2139    | 1.2099    |
| X8,(2) | 1.24798 | 1.5680 | 1.4747 | 1.4312 | 1.5544 | 1.5425     | 1.5257     | 1.5150     | 1.4856    | 1.4765    |
| X8,(3) | 1.57317 | 1.9919 | 1.8772 | 1.7975 | 1.9682 | 1.9471     | 1.9169     | 1.8978     | 1.8448    | 1.8284    |
| X8,(4) | 2.03150 | 2.6277 | 2.4696 | 2.3381 | 2.5889 | 2.5522     | 2.4997     | 2.4664     | 2.3740    | 2.3455    |
| X8,(5) | 2.81503 | 3.8994 | 3.5957 | 3.3723 | 3.8249 | 3.7488     | 3.6399     | 3.5702     | 3.3858    | 3.3263    |

Table 3: Different Point Predictors for Rayleigh Distribution

|        | MLP    | BUP    | CMP    | ADP    | SEP    | GEP        |            |            |           |           |
|--------|--------|--------|--------|--------|--------|------------|------------|------------|-----------|-----------|
|        |        |        |        |        |        | $q = -0.8$ | $q = -0.5$ | $q = -0.3$ | $q = 0.3$ | $q = 0.5$ |
| X3,(1) | 0.4881 | 0.6768 | 0.6383 | 0.6343 | 1.4695 | 0.6766     | 0.6710     | 0.6675     | 0.6574    | 0.6542    |
| X3,(2) | 0.7316 | 0.8940 | 0.8637 | 0.8547 | 1.1111 | 0.8934     | 0.8838     | 0.8775     | 0.8593    | 0.8535    |
| X3,(3) | 1.0213 | 1.2163 | 1.1809 | 1.1671 | 0.8166 | 1.2140     | 1.1985     | 1.1884     | 1.1587    | 1.1491    |
| X5,(1) | 0.2036 | 0.4943 | 0.4590 | 0.4534 | 2.0101 | 0.4886     | 0.4756     | 0.4672     | 0.4430    | 0.4354    |
| X5,(2) | 0.5818 | 0.7651 | 0.7411 | 0.7306 | 1.2987 | 0.7595     | 0.7439     | 0.7335     | 0.7026    | 0.6924    |
| X5,(3) | 0.9199 | 1.1251 | 1.0944 | 1.0795 | 0.8833 | 1.1187     | 1.0988     | 1.0855     | 1.0459    | 1.0328    |
| X8,(1) | 0.7692 | 0.8555 | 0.8326 | 0.8308 | 1.1655 | 0.8571     | 0.8559     | 0.8551     | 0.8527    | 0.8520    |
| X8,(2) | 0.8690 | 0.9529 | 0.9314 | 0.9269 | 1.0444 | 0.9557     | 0.9531     | 0.9514     | 0.9466    | 0.9450    |
| X8,(3) | 0.9828 | 1.0692 | 1.0485 | 1.0411 | 0.9298 | 1.0727     | 1.0686     | 1.0659     | 1.0582    | 1.0557    |
| X8,(4) | 1.1238 | 1.2226 | 1.2003 | 1.1899 | 0.8126 | 1.2264     | 1.2201     | 1.2160     | 1.2042    | 1.2004    |
| X8,(5) | 1.3306 | 1.4788 | 1.4454 | 1.4321 | 0.6716 | 1.4816     | 1.4709     | 1.4639     | 1.4436    | 1.4371    |

## 4.2 Real data analysis

The dataset contains the number of revolutions before failure for each of the 23 ball bearings in a life test, as reported by Lawless (1982, p. 288). The following progressive type-II censored sample data were recently analysed by Raqab and Madi (2011).

|       |       |       |       |       |       |       |
|-------|-------|-------|-------|-------|-------|-------|
| i     | 1     | 2     | 3     | 4     | 5     | 6     |
| $y_i$ | 17.88 | 28.92 | 33.00 | 41.52 | 51.84 | 51.96 |
| $R_i$ | 0     | 0     | 0     | 3     | 0     | 0     |
| i     | 7     | 8     | 9     | 10    | 11    | 12    |
| $y_i$ | 54.12 | 55.56 | 68.88 | 84.12 | 93.12 | 98.64 |
| $R_i$ | 0     | 3     | 0     | 0     | 0     | 5     |

We have fitted this data to the Rayleigh distribution with parameter  $\lambda$ . The MLE,  $\hat{\lambda} = 18112.6$ .

We estimate the predictors MLP, BUP, CMP, ADP, SEP and BGP. Since we have no prior information, to compute Bayesian predictors, we assume a gamma prior with  $\alpha = 0$  and  $\beta = 0$ , i.e., an improper prior. The estimates are shown in Table 4.

Table 4: Estimated Predictors based on the dataset

| Term    | MLP      | BUP      | CMP      | ADP      | SEP      | GEP        |            |            |           |           |
|---------|----------|----------|----------|----------|----------|------------|------------|------------|-----------|-----------|
|         |          |          |          |          |          | $q = -0.8$ | $q = -0.5$ | $q = -0.3$ | $q = 0.3$ | $q = 0.5$ |
| X4,(1)  | 41.5200  | 66.1024  | 61.7767  | 62.2734  | 67.4923  | 66.8744    | 65.9794    | 65.4040    | 63.7792   | 63.2708   |
| X4,(2)  | 73.4570  | 92.5938  | 89.4498  | 90.4745  | 95.1209  | 94.2162    | 92.8786    | 92.0003    | 89.4342   | 88.6030   |
| X4,(3)  | 108.0429 | 130.2884 | 126.5647 | 128.3401 | 134.2126 | 132.9122   | 130.9785   | 129.7011   | 125.9279  | 124.6908  |
| X8,(1)  | 55.5600  | 76.2316  | 71.9678  | 72.3946  | 77.5184  | 77.0860    | 76.4608    | 76.0592    | 74.9246   | 74.5688   |
| X8,(2)  | 82.2127  | 100.1572 | 96.7691  | 97.7171  | 102.5872 | 101.8598   | 100.7906   | 100.0926   | 98.0703   | 97.4202   |
| X8,(3)  | 114.1765 | 135.8016 | 131.8394 | 133.5447 | 139.6553 | 138.5024   | 136.7964   | 135.6751   | 132.3895  | 131.3211  |
| X12,(1) | 98.6400  | 107.1353 | 104.8109 | 104.9871 | 107.8014 | 107.7275   | 107.6192   | 107.5486   | 107.3445  | 107.2789  |
| X12,(2) | 108.4007 | 116.9087 | 114.6319 | 115.0622 | 118.2423 | 118.0868   | 117.8585   | 117.7096   | 117.2773  | 117.1380  |
| X12,(3) | 119.8167 | 128.7873 | 126.5196 | 127.2446 | 130.8322 | 130.5772   | 130.2023   | 129.9572   | 129.2446  | 129.0143  |
| X12,(4) | 134.2686 | 144.7227 | 142.2175 | 143.3362 | 147.6064 | 147.2071   | 146.6193   | 146.2346   | 145.1143  | 144.7518  |
| X12,(5) | 155.9019 | 171.8304 | 168.0629 | 169.9106 | 175.9576 | 175.2448   | 174.1958   | 173.5097   | 171.5136  | 170.8684  |

## 5 Concluding remarks

This project investigates the prediction of failure times for units withdrawn at different stages under a progressively Type-II censored sampling scheme arising from proportional hazard rate (PHR) models. Several prediction procedures, including the maximum likelihood predictor (MLP), best unbiased predictor (BUP), conditional median predictor (CMP), and Bayesian predictors, have been examined. The study considers three important baseline distributions, namely the exponential, Lindley, and Rayleigh models, in order to assess and compare their predictive performance.

The predictors are evaluated through their bias and mean squared prediction error (MSPE). Based on numerical and simulation results, the BUP emerges as the most effective predictor within the frequentist framework, whereas the squared-error-loss-based predictor (SEP) demonstrates superior performance among the Bayesian approaches. A comparison of the underlying baseline models further indicates that the Rayleigh-based PHR model generally provides the most satisfactory predictive results.

An interesting extension of the present work would be to incorporate covariate information into the hazard structure by expressing the propor-

tionality parameter as

$$\theta = \beta z$$

or

$$\theta = \exp(\beta z),$$

where  $z = (z_1, z_2, \dots, z_k)$  denotes a vector of explanatory variables and  $\beta = (\beta_1, \beta_2, \dots, \beta_k)$  represents a vector of unknown parameters. The resulting prediction problem becomes considerably more complex because both the regression parameters and the covariate effects influence the prediction mechanism. Consequently, further theoretical and computational developments are required to address this setting effectively.

It is also worth noting that the proposed methodology encompasses several commonly used censoring schemes as special cases. In particular, the results reduce to the conventional Type-II censoring scheme when

$$R_1 = R_2 = \dots = R_{m-1} = 0 \quad \text{and} \quad R_m = n - m,$$

and to the complete sample case when

$$n = m \quad \text{and} \quad R_i = 0, \quad i = 1, 2, \dots, m.$$

Therefore, the procedures developed in this study have broad applicability in reliability and survival analysis.

Table 5: Bias and MSPE for Exponential Distribution ( $\theta = 0.75$ )

|        | MLP     | BUP     | CMP     | ADP     | SEP     | GEP        |            |            |           |           |
|--------|---------|---------|---------|---------|---------|------------|------------|------------|-----------|-----------|
|        |         |         |         |         |         | $q = -0.8$ | $q = -0.5$ | $q = -0.3$ | $q = 0.3$ | $q = 0.5$ |
| X3,(1) | -0.0579 | 0.0030  | -0.1335 | -0.1625 | -0.0094 | -0.0362    | -0.0731    | -0.0956    | -0.1532   | -0.1694   |
| X3,(2) | -0.1389 | 0.0509  | -0.1389 | -0.2282 | 0.0182  | -0.0339    | -0.1080    | -0.1549    | -0.2831   | -0.3218   |
| X3,(3) | -0.6727 | -0.0700 | -0.4137 | -0.6107 | -0.1410 | -0.2447    | -0.3934    | -0.4883    | -0.7523   | -0.8336   |
| X5,(1) | -0.0695 | -0.0075 | -0.1463 | -0.1762 | -0.0207 | -0.0417    | -0.0704    | -0.0879    | -0.1334   | -0.1465   |
| X5,(2) | -0.1442 | 0.0442  | -0.1442 | -0.2325 | 0.0123  | -0.0323    | -0.0953    | -0.1348    | -0.2418   | -0.2739   |
| X5,(3) | -0.6247 | -0.0378 | -0.3725 | -0.5600 | -0.1016 | -0.1951    | -0.3283    | -0.4126    | -0.6446   | -0.7152   |
| X8,(1) | -0.0254 | -0.0018 | -0.0838 | -0.1012 | -0.0092 | -0.0149    | -0.0228    | -0.0278    | -0.0415   | -0.0456   |
| X8,(2) | -0.0496 | 0.0101  | -0.0882 | -0.1361 | -0.0069 | -0.0196    | -0.0375    | -0.0489    | -0.0803   | -0.0899   |
| X8,(3) | -0.1283 | -0.0097 | -0.1283 | -0.2132 | -0.0371 | -0.0593    | -0.0909    | -0.1109    | -0.1663   | -0.1834   |
| X8,(4) | -0.2383 | 0.0018  | -0.1599 | -0.2969 | -0.0409 | -0.0789    | -0.1330    | -0.1673    | -0.2627   | -0.2921   |
| X8,(5) | -0.6882 | -0.0282 | -0.3487 | -0.5966 | -0.1151 | -0.1965    | -0.3123    | -0.3856    | -0.5877   | -0.6498   |

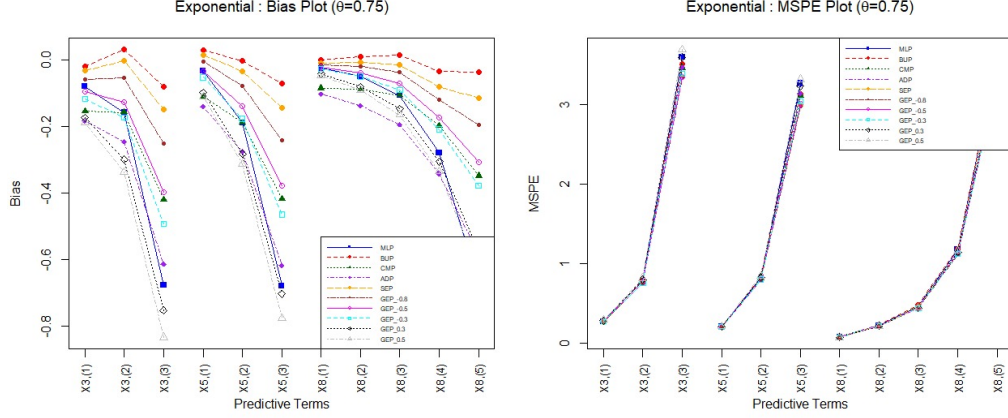


Figure 3: Bias and MSPE plots for Exponential Distribution ( $\theta = 0.75$ )

Table 6: Bias and MSPE for Exponential Distribution ( $\theta = 1$ )

|        |      | MLP      | BUP      | CMP      | ADP      | SEP      | GEP        |            |            |           |           |
|--------|------|----------|----------|----------|----------|----------|------------|------------|------------|-----------|-----------|
|        |      |          |          |          |          |          | $q = -0.8$ | $q = -0.5$ | $q = -0.3$ | $q = 0.3$ | $q = 0.5$ |
| X3,(1) | Bias | -0.04343 | 0.00225  | -0.10011 | -0.11587 | 0.00222  | -0.01864   | -0.04737   | -0.06485   | -0.10966  | -0.12226  |
|        | MSPE | 0.12669  | 0.12847  | 0.13098  | 0.13215  | 0.12546  | 0.12446    | 0.12466    | 0.12568    | 0.13141   | 0.13381   |
| X3,(2) | Bias | -0.10414 | 0.03817  | -0.10414 | -0.15319 | 0.03677  | -0.00352   | -0.06100   | -0.09733   | -0.19682  | -0.22682  |
|        | MSPE | 0.42968  | 0.44378  | 0.42968  | 0.42625  | 0.42761  | 0.42092    | 0.41757    | 0.41921    | 0.43861   | 0.44874   |
| X3,(3) | Bias | -0.50452 | -0.05247 | -0.31023 | -0.41704 | -0.05477 | -0.13491   | -0.24997   | -0.32332   | -0.52764  | -0.59060  |
|        | MSPE | 1.95008  | 1.88280  | 1.86457  | 1.84355  | 1.79234  | 1.77914    | 1.78499    | 1.80397    | 1.91955   | 1.97377   |
| X5,(1) | Bias | -0.05210 | -0.00564 | -0.10974 | -0.12617 | -0.00630 | -0.02262   | -0.04504   | -0.05869   | -0.09416  | -0.10436  |
|        | MSPE | 0.12890  | 0.13002  | 0.13423  | 0.13578  | 0.12691  | 0.12630    | 0.12641    | 0.12702    | 0.13050   | 0.13201   |
| X5,(2) | Bias | -0.10818 | 0.03317  | -0.10818 | -0.15642 | 0.03240  | -0.00226   | -0.05124   | -0.08195   | -0.16519  | -0.19013  |
|        | MSPE | 0.38704  | 0.40312  | 0.38704  | 0.38105  | 0.38489  | 0.37830    | 0.37350    | 0.37317    | 0.38273   | 0.38856   |
| X5,(3) | Bias | -0.46854 | -0.02833 | -0.27934 | -0.37906 | -0.02522 | -0.09767   | -0.20096   | -0.26634   | -0.44636  | -0.50120  |
|        | MSPE | 1.76424  | 1.72074  | 1.69213  | 1.66358  | 1.63469  | 1.61882    | 1.61616    | 1.62662    | 1.70413   | 1.74193   |
| X8,(1) | Bias | -0.01904 | -0.00133 | -0.06282 | -0.07230 | -0.00137 | -0.00580   | -0.01204   | -0.01596   | -0.02669  | -0.02996  |
|        | MSPE | 0.04073  | 0.04113  | 0.04278  | 0.04341  | 0.04019  | 0.04009    | 0.04002    | 0.04002    | 0.04019   | 0.04028   |
| X8,(2) | Bias | -0.03717 | 0.00759  | -0.06617 | -0.09232 | 0.00734  | -0.00259   | -0.01669   | -0.02559   | -0.05013  | -0.05766  |
|        | MSPE | 0.12002  | 0.12343  | 0.12022  | 0.11884  | 0.11815  | 0.11728    | 0.11642    | 0.11610    | 0.11611   | 0.11638   |
| X8,(3) | Bias | -0.09623 | -0.00726 | -0.09623 | -0.14193 | -0.00609 | -0.02342   | -0.04811   | -0.06375   | -0.10708  | -0.12042  |
|        | MSPE | 0.27638  | 0.28311  | 0.27638  | 0.26918  | 0.26769  | 0.26576    | 0.26415    | 0.26381    | 0.26565   | 0.26704   |
| X8,(4) | Bias | -0.17872 | 0.00134  | -0.11989 | -0.19262 | 0.00494  | -0.02463   | -0.06684   | -0.09362   | -0.16796  | -0.19090  |
|        | MSPE | 0.59102  | 0.60797  | 0.58854  | 0.56497  | 0.56955  | 0.56359    | 0.55840    | 0.55712    | 0.56178   | 0.56566   |
| X8,(5) | Bias | -0.51613 | -0.02095 | -0.26156 | -0.39424 | -0.02282 | -0.08602   | -0.17597   | -0.23288   | -0.39001  | -0.43821  |
|        | MSPE | 2.29568  | 2.28782  | 2.22278  | 2.14404  | 2.14699  | 2.12575    | 2.11060    | 2.11017    | 2.14593   | 2.16771   |

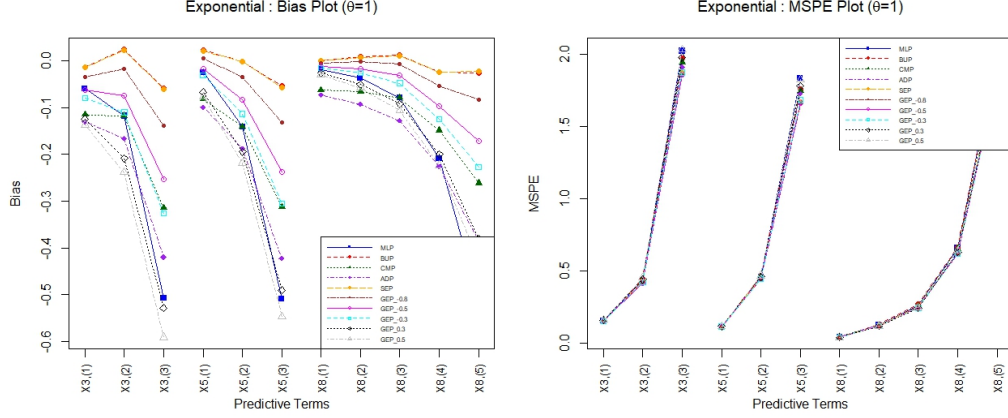


Figure 4: Bias and MSPE plots for Exponential Distribution ( $\theta = 1$ )

Table 7: Bias and MSPE for Exponential Distribution ( $\theta = 2$ )

|        |      | MLP      | BUP        | CMP      | ADP      | SEP     | GEP        |            |            |           |           |
|--------|------|----------|------------|----------|----------|---------|------------|------------|------------|-----------|-----------|
|        |      |          |            |          |          |         | $q = -0.8$ | $q = -0.5$ | $q = -0.3$ | $q = 0.3$ | $q = 0.5$ |
| X3,(1) | Bias | -0.02172 | 0.0011267  | -0.05006 | -0.04597 | 0.01963 | 0.007688   | -0.008775  | -0.018794  | -0.044463 | -0.051658 |
|        | MSPE | 0.03167  | 0.0321165  | 0.03275  | 0.03180  | 0.03175 | 0.031088   | 0.030687   | 0.030731   | 0.031841  | 0.032407  |
| X3,(2) | Bias | -0.05207 | 0.0190842  | -0.05207 | -0.04071 | 0.06468 | 0.041954   | 0.009491   | -0.011058  | -0.067445 | -0.084470 |
|        | MSPE | 0.10742  | 0.1109437  | 0.10742  | 0.10235  | 0.11075 | 0.106991   | 0.103563   | 0.102571   | 0.104552  | 0.106505  |
| X3,(3) | Bias | -0.25226 | -0.0262349 | -0.15512 | -0.12660 | 0.07448 | 0.029618   | -0.034852  | -0.075989  | -0.190758 | -0.226185 |
|        | MSPE | 0.48752  | 0.4707043  | 0.46614  | 0.43343  | 0.45289 | 0.441130   | 0.431853   | 0.430648   | 0.446706  | 0.457440  |
| X5,(1) | Bias | -0.02605 | -0.0028198 | -0.05487 | -0.05112 | 0.01537 | 0.005907   | -0.007076  | -0.014976  | -0.035441 | -0.041305 |
|        | MSPE | 0.03223  | 0.0325046  | 0.03356  | 0.03258  | 0.03195 | 0.031483   | 0.031150   | 0.031124   | 0.031676  | 0.031997  |
| X5,(2) | Bias | -0.05409 | 0.0165844  | -0.05409 | -0.04233 | 0.06250 | 0.042732   | 0.014765   | -0.002783  | -0.050382 | -0.064642 |
|        | MSPE | 0.09676  | 0.1007791  | 0.09676  | 0.09094  | 0.09986 | 0.096403   | 0.092941   | 0.091630   | 0.091414  | 0.092299  |
| X5,(3) | Bias | -0.23427 | -0.0141658 | -0.13967 | -0.10761 | 0.08924 | 0.048324   | -0.010095  | -0.047102  | -0.149180 | -0.180320 |
|        | MSPE | 0.44106  | 0.4301848  | 0.42303  | 0.39155  | 0.41649 | 0.404673   | 0.394072   | 0.391179   | 0.398558  | 0.405297  |
| X8,(1) | Bias | -0.00952 | -0.0006643 | -0.03141 | -0.02897 | 0.01043 | 0.007779   | 0.004056   | 0.001725   | -0.004628 | -0.006556 |
|        | MSPE | 0.01018  | 0.0102824  | 0.01069  | 0.01039  | 0.01016 | 0.010076   | 0.009988   | 0.009948   | 0.009895  | 0.009896  |
| X8,(2) | Bias | -0.01858 | 0.0037929  | -0.03309 | -0.02667 | 0.02867 | 0.022816   | 0.014518   | 0.009289   | -0.005096 | -0.009496 |
|        | MSPE | 0.03001  | 0.0308571  | 0.03005  | 0.02829  | 0.03035 | 0.029842   | 0.029253   | 0.028956   | 0.028438  | 0.028366  |
| X8,(3) | Bias | -0.04811 | -0.0036301 | -0.04811 | -0.03508 | 0.04048 | 0.030365   | 0.015968   | 0.006853   | -0.018371 | -0.026129 |
|        | MSPE | 0.06910  | 0.0707771  | 0.06910  | 0.06349  | 0.06855 | 0.067231   | 0.065726   | 0.065002   | 0.063915  | 0.063851  |
| X8,(4) | Bias | -0.08936 | 0.0006685  | -0.05994 | -0.03625 | 0.07376 | 0.056683   | 0.032300   | 0.016830   | -0.026118 | -0.039361 |
|        | MSPE | 0.14775  | 0.1519913  | 0.14714  | 0.13328  | 0.14782 | 0.143966   | 0.139544   | 0.137397   | 0.134120  | 0.133904  |
| X8,(5) | Bias | -0.25806 | -0.0104635 | -0.13078 | -0.09077 | 0.11545 | 0.079509   | 0.028311   | -0.004081  | -0.093598 | -0.121061 |
|        | MSPE | 0.57392  | 0.5719932  | 0.55569  | 0.50539  | 0.54997 | 0.535942   | 0.520762   | 0.514077   | 0.507316  | 0.508694  |

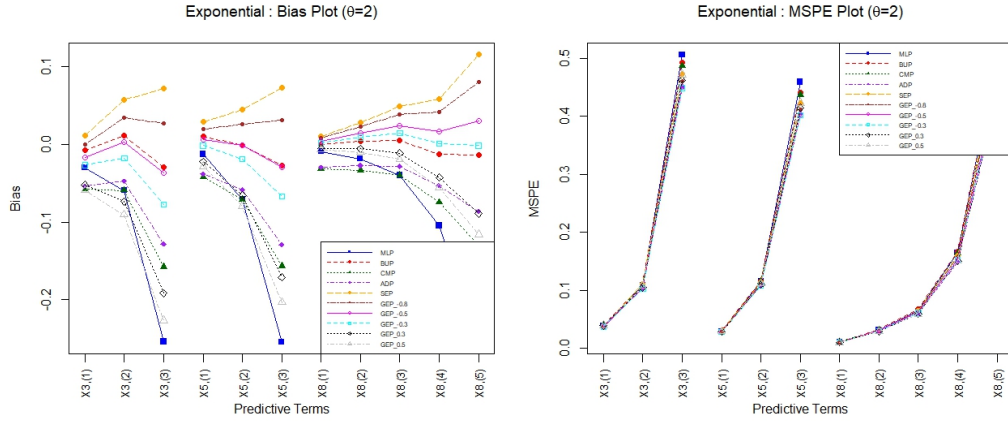


Figure 5: Bias and MSPE plots for Exponential Distribution ( $\theta = 2$ )

Table 8: Bias and MSPE for Lindley Distribution( $\theta = 0.75$ )

| Term   | Metric | MLP     | BUP     | CMP     | ADP     | SEP     | GEP <sub>-0.8</sub> | GEP <sub>-0.5</sub> | GEP <sub>-0.3</sub> | GEP <sub>0.3</sub> | GEP <sub>0.5</sub> |
|--------|--------|---------|---------|---------|---------|---------|---------------------|---------------------|---------------------|--------------------|--------------------|
| X3,(1) | Bias   | -0.0707 | -0.0078 | -0.1822 | -0.2242 | -0.0263 | -0.0598             | -0.1067             | -0.1357             | -0.2125            | -0.2347            |
|        | MSPE   | 0.4025  | 0.4054  | 0.4183  | 0.4274  | 0.3960  | 0.3958              | 0.3996              | 0.4044              | 0.4261             | 0.4348             |
| X3,(2) | Bias   | -0.2037 | 0.0050  | -0.2037 | -0.3184 | -0.0402 | -0.0988             | -0.1834             | -0.2376             | -0.3897            | -0.4368            |
|        | MSPE   | 1.2384  | 1.2575  | 1.2384  | 1.2423  | 1.2062  | 1.2006              | 1.2058              | 1.2173              | 1.2840             | 1.3149             |
| X3,(3) | Bias   | -0.6756 | -0.0018 | -0.3585 | -0.5940 | -0.0956 | -0.2044             | -0.3620             | -0.4634             | -0.7507            | -0.8410            |
|        | MSPE   | 4.3921  | 4.3105  | 4.2232  | 4.2176  | 4.1089  | 4.0885              | 4.1051              | 4.1445              | 4.3792             | 4.4901             |
| X5,(1) | Bias   | -0.0743 | -0.0065 | -0.1837 | -0.2259 | -0.0260 | -0.0509             | -0.0857             | -0.1072             | -0.1645            | -0.1813            |
|        | MSPE   | 0.4361  | 0.4395  | 0.4522  | 0.4618  | 0.4302  | 0.4297              | 0.4313              | 0.4335              | 0.4445             | 0.4491             |
| X5,(2) | Bias   | -0.2711 | -0.0597 | -0.2711 | -0.3840 | -0.1036 | -0.1528             | -0.2231             | -0.2676             | -0.3909            | -0.4286            |
|        | MSPE   | 1.4834  | 1.4757  | 1.4834  | 1.5030  | 1.4309  | 1.4319              | 1.4425              | 1.4549              | 1.5116             | 1.5356             |
| X5,(3) | Bias   | -0.6940 | -0.0223 | -0.3811 | -0.6133 | -0.1132 | -0.2120             | -0.3540             | -0.4447             | -0.6981            | -0.7766            |
|        | MSPE   | 4.8813  | 4.7959  | 4.7114  | 4.7021  | 4.5906  | 4.5727              | 4.5844              | 4.6149              | 4.7963             | 4.8809             |
| X8,(1) | Bias   | -0.0314 | -0.0042 | -0.1108 | -0.1346 | -0.0149 | -0.0214             | -0.0307             | -0.0366             | -0.0529            | -0.0579            |
|        | MSPE   | 0.1379  | 0.1389  | 0.1444  | 0.1479  | 0.1361  | 0.1360              | 0.1360              | 0.1361              | 0.1368             | 0.1371             |
| X8,(2) | Bias   | -0.1174 | -0.0500 | -0.1670 | -0.2273 | -0.0719 | -0.0859             | -0.1058             | -0.1185             | -0.1540            | -0.1650            |
|        | MSPE   | 0.4212  | 0.4209  | 0.4280  | 0.4360  | 0.4086  | 0.4091              | 0.4105              | 0.4118              | 0.4173             | 0.4196             |
| X8,(3) | Bias   | -0.1365 | 0.0002  | -0.1365 | -0.2438 | -0.0379 | -0.0619             | -0.0963             | -0.1182             | -0.1798            | -0.1990            |
|        | MSPE   | 0.7318  | 0.7500  | 0.7318  | 0.7240  | 0.7094  | 0.7068              | 0.7052              | 0.7056              | 0.7121             | 0.7157             |
| X8,(4) | Bias   | -0.3127 | -0.0334 | -0.2148 | -0.3880 | -0.0961 | -0.1365             | -0.1946             | -0.2317             | -0.3359            | -0.3685            |
|        | MSPE   | 1.8409  | 1.8665  | 1.8292  | 1.8013  | 1.7652  | 1.7597              | 1.7582              | 1.7610              | 1.7850             | 1.7973             |
| X8,(5) | Bias   | -0.9593 | -0.2326 | -0.5673 | -0.8466 | -0.3372 | -0.4183             | -0.5345             | -0.6085             | -0.8149            | -0.8790            |
|        | MSPE   | 5.9428  | 5.5676  | 5.5894  | 5.6412  | 5.3311  | 5.3450              | 5.3899              | 5.4339              | 5.6208             | 5.6975             |

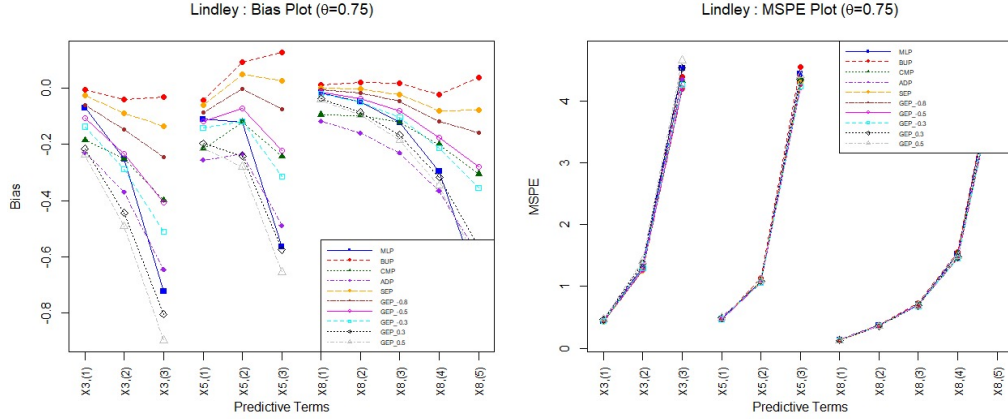


Figure 6: Bias and MSPE plots for Lindley Distribution ( $\theta = 0.75$ )

Table 9: Bias and MSPE Values for Lindley Distribution( $\theta = 1$ )

| Term   | Metric | MLP      | BUP      | CMP      | ADP      | SEP      | GEP <sub>-0.8</sub> | GEP <sub>-0.5</sub> | GEP <sub>-0.3</sub> | GEP <sub>0.3</sub> | GEP <sub>0.5</sub> |
|--------|--------|----------|----------|----------|----------|----------|---------------------|---------------------|---------------------|--------------------|--------------------|
| X3,(1) | Bias   | -0.06789 | -0.01741 | -0.15651 | -0.18089 | -0.01944 | -0.04705            | -0.08575            | -0.10973            | -0.17306           | -0.19141           |
|        | MSPE   | 0.28125  | 0.28120  | 0.29425  | 0.29812  | 0.27554  | 0.27558             | 0.27844             | 0.28186             | 0.29691            | 0.30290            |
| X3,(2) | Bias   | -0.12608 | 0.03804  | -0.12608 | -0.19193 | 0.03221  | -0.01557            | -0.08469            | -0.12905            | -0.25361           | -0.29222           |
|        | MSPE   | 0.76082  | 0.78750  | 0.76081  | 0.74168  | 0.74924  | 0.73917             | 0.73342             | 0.73527             | 0.76350            | 0.77917            |
| X3,(3) | Bias   | -0.47995 | 0.04627  | -0.23038 | -0.36422 | 0.03264  | -0.05460            | -0.18116            | -0.26273            | -0.49434           | -0.56727           |
|        | MSPE   | 2.77851  | 2.79365  | 2.70603  | 2.62999  | 2.65044  | 2.61909             | 2.60313             | 2.61137             | 2.71389            | 2.77037            |
| X5,(1) | Bias   | -0.04152 | 0.01078  | -0.12612 | -0.14926 | 0.00933  | -0.01085            | -0.03898            | -0.05639            | -0.10268           | -0.11631           |
|        | MSPE   | 0.23266  | 0.23477  | 0.24173  | 0.24499  | 0.23023  | 0.22922             | 0.22928             | 0.23017             | 0.23572            | 0.23823            |
| X5,(2) | Bias   | -0.17030 | -0.00733 | -0.17030 | -0.23226 | -0.01032 | -0.04991            | -0.10658            | -0.14257            | -0.24221           | -0.27274           |
|        | MSPE   | 0.80243  | 0.81199  | 0.80243  | 0.79148  | 0.77804  | 0.77331             | 0.77247             | 0.77555             | 0.79874            | 0.81014            |
| X5,(3) | Bias   | -0.51732 | -0.00908 | -0.27842 | -0.39910 | -0.01219 | -0.08961            | -0.20109            | -0.27238            | -0.47221           | -0.53429           |
|        | MSPE   | 2.49177  | 2.44058  | 2.39381  | 2.33888  | 2.31474  | 2.29538             | 2.29045             | 2.30149             | 2.39145            | 2.43705            |
| X8,(1) | Bias   | -0.02585 | -0.00466 | -0.08799 | -0.10157 | -0.00553 | -0.01084            | -0.01841            | -0.02320            | -0.03650           | -0.04060           |
|        | MSPE   | 0.09488  | 0.09563  | 0.09841  | 0.09919  | 0.09334  | 0.09317             | 0.09303             | 0.09301             | 0.09319            | 0.09332            |
| X8,(2) | Bias   | -0.05187 | 0.00132  | -0.09137 | -0.12675 | -0.00067 | -0.01219            | -0.02870            | -0.03920            | -0.06854           | -0.07765           |
|        | MSPE   | 0.20778  | 0.21057  | 0.20974  | 0.20938  | 0.20264  | 0.20189             | 0.20132             | 0.20125             | 0.20231            | 0.20301            |
| X8,(3) | Bias   | -0.12728 | -0.02146 | -0.12728 | -0.18730 | -0.02351 | -0.04282            | -0.07058            | -0.08831            | -0.13806           | -0.15358           |
|        | MSPE   | 0.44995  | 0.45407  | 0.44995  | 0.44181  | 0.43042  | 0.42914             | 0.42869             | 0.42926             | 0.43445            | 0.43715            |
| X8,(4) | Bias   | -0.18644 | 0.02869  | -0.11040 | -0.20580 | 0.02449  | -0.00803            | -0.05482            | -0.08474            | -0.16887           | -0.19514           |
|        | MSPE   | 0.95762  | 0.99683  | 0.95919  | 0.90795  | 0.92809  | 0.91874             | 0.90932             | 0.90579             | 0.90623            | 0.90951            |
| X8,(5) | Bias   | -0.55223 | 0.00261  | -0.25095 | -0.39834 | -0.00107 | -0.06517            | -0.15709            | -0.21570            | -0.37955           | -0.43046           |
|        | MSPE   | 2.67934  | 2.67838  | 2.59097  | 2.46583  | 2.48757  | 2.46232             | 2.44171             | 2.43816             | 2.46788            | 2.48895            |



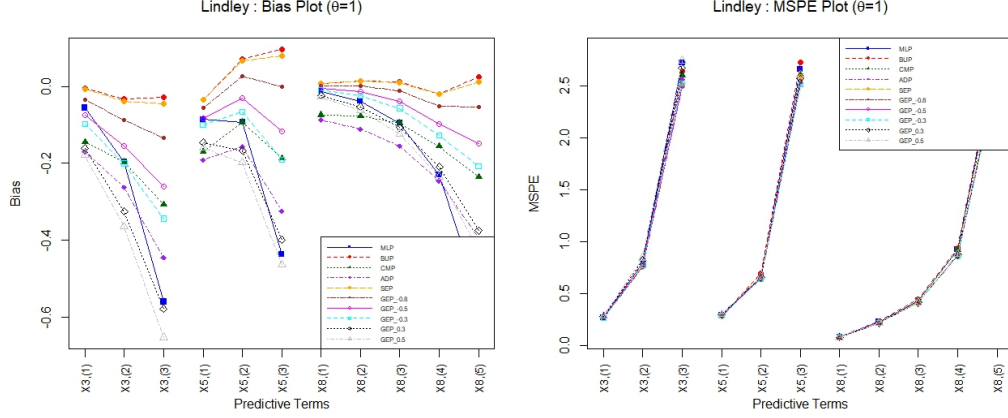


Figure 7: Bias and MSPE plots for Lindley Distribution ( $\theta = 1$ )

Table 10: Bias and MSPE for Lindley Distribution( $\theta = 2$ )

| Term   | Metric | MLP       | BUP        | CMP      | ADP      | SEP      | GEP <sub>-0.8</sub> | GEP <sub>-0.5</sub> | GEP <sub>-0.3</sub> | GEP <sub>0.3</sub> | GEP <sub>0.5</sub> |
|--------|--------|-----------|------------|----------|----------|----------|---------------------|---------------------|---------------------|--------------------|--------------------|
| X3,(1) | Bias   | -0.035016 | -0.0062572 | -0.08253 | -0.07435 | 0.022930 | 0.005452            | -0.019057           | -0.034220           | -0.074089          | -0.085556          |
|        | MSPE   | 0.067653  | 0.0678244  | 0.07095  | 0.06815  | 0.066305 | 0.065146            | 0.064606            | 0.064907            | 0.068010           | 0.069525           |
| X3,(2) | Bias   | -0.062982 | 0.0286855  | -0.06297 | -0.04323 | 0.093274 | 0.063341            | 0.019931            | -0.007965           | -0.086554          | -0.110930          |
|        | MSPE   | 0.203465  | 0.2121923  | 0.20346  | 0.18880  | 0.207754 | 0.200492            | 0.193362            | 0.190899            | 0.192940           | 0.196256           |
| X3,(3) | Bias   | -0.242279 | 0.0447992  | -0.10426 | -0.06136 | 0.173399 | 0.120230            | 0.042780            | -0.007342           | -0.150553          | -0.195908          |
|        | MSPE   | 0.807569  | 0.8349229  | 0.79782  | 0.72902  | 0.806255 | 0.779337            | 0.750965            | 0.739453            | 0.736198           | 0.744320           |
| X5,(1) | Bias   | -0.009905 | 0.0198393  | -0.05675 | -0.04972 | 0.047582 | 0.034375            | 0.015983            | 0.004626            | -0.025444          | -0.034247          |
|        | MSPE   | 0.061408  | 0.0632295  | 0.06228  | 0.06000  | 0.063006 | 0.061532            | 0.060094            | 0.059562            | 0.059458           | 0.059783           |
| X5,(2) | Bias   | -0.097400 | -0.0062880 | -0.09740 | -0.07738 | 0.058195 | 0.032893            | -0.003416           | -0.026522           | -0.090633          | -0.110295          |
|        | MSPE   | 0.255099  | 0.2569089  | 0.25510  | 0.23997  | 0.248700 | 0.244219            | 0.240155            | 0.239018            | 0.241766           | 0.244345           |
| X5,(3) | Bias   | -0.360951 | -0.0866770 | -0.23029 | -0.17904 | 0.048506 | 0.001461            | -0.066524           | -0.110169           | -0.233142          | -0.271517          |
|        | MSPE   | 0.861375  | 0.7909575  | 0.80723  | 0.74913  | 0.750303 | 0.741609            | 0.737273            | 0.739712            | 0.768032           | 0.783403           |
| X8,(1) | Bias   | -0.012229 | -0.0007777 | -0.04567 | -0.04132 | 0.016101 | 0.012610            | 0.007654            | 0.004525            | -0.004110          | -0.006761          |
|        | MSPE   | 0.024822  | 0.0250645  | 0.02573  | 0.02481  | 0.024566 | 0.024406            | 0.024221            | 0.024129            | 0.023980           | 0.023964           |
| X8,(2) | Bias   | -0.056756 | -0.0282040 | -0.07805 | -0.06648 | 0.008484 | 0.001035            | -0.009617           | -0.016389           | -0.035260          | -0.041105          |
|        | MSPE   | 0.072829  | 0.0722065  | 0.07442  | 0.06967  | 0.068439 | 0.068101            | 0.067817            | 0.067759            | 0.068093           | 0.068345           |
| X8,(3) | Bias   | -0.050406 | 0.0068413  | -0.05041 | -0.02949 | 0.068212 | 0.055803            | 0.037970            | 0.026574            | -0.005402          | -0.015371          |
|        | MSPE   | 0.142608  | 0.1469264  | 0.14261  | 0.13029  | 0.141955 | 0.139515            | 0.136572            | 0.135040            | 0.132188           | 0.131734           |
| X8,(4) | Bias   | -0.101697 | 0.0169140  | -0.05929 | -0.02898 | 0.109518 | 0.088982            | 0.059391            | 0.040436            | -0.012942          | -0.029641          |
|        | MSPE   | 0.294099  | 0.3050624  | 0.29441  | 0.26554  | 0.293755 | 0.287354            | 0.279694            | 0.275758            | 0.268730           | 0.267760           |
| X8,(5) | Bias   | -0.396712 | -0.0938239 | -0.23014 | -0.17091 | 0.064209 | 0.024810            | -0.031833           | -0.068041           | -0.169637          | -0.201302          |
|        | MSPE   | 1.130376  | 1.0805242  | 1.07705  | 0.97315  | 1.004378 | 0.991803            | 0.979537            | 0.975302            | 0.978391           | 0.983860           |

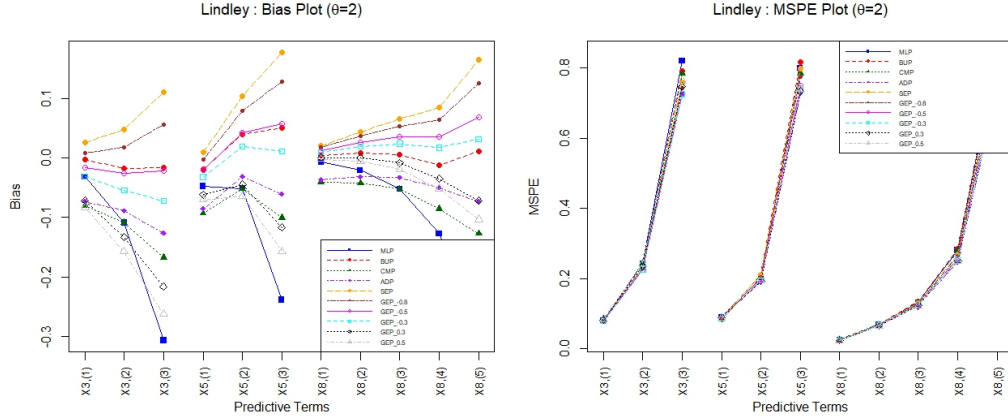


Figure 8: Bias and MSPE plots for Lindley Distribution ( $\theta = 2$ )

Table 11: Bias and MSPE for Rayleigh Distribution( $\theta = 0.75$ )

| Term   | Metric | MLP     | BUP     | CMP     | ADP     | SEP     | GEP <sub>-0.8</sub> | GEP <sub>-0.5</sub> | GEP <sub>-0.3</sub> | GEP <sub>0.3</sub> | GEP <sub>0.5</sub> |
|--------|--------|---------|---------|---------|---------|---------|---------------------|---------------------|---------------------|--------------------|--------------------|
| X3,(1) | Bias   | -0.0152 | -0.0144 | -0.0651 | -0.0842 | -0.0243 | -0.0319             | -0.0429             | -0.0500             | -0.0700            | -0.0763            |
|        | MSPE   | 0.0692  | 0.0691  | 0.0719  | 0.0735  | 0.0682  | 0.0685              | 0.0691              | 0.0697              | 0.0718             | 0.0727             |
| X3,(2) | Bias   | -0.0344 | 0.0021  | -0.0344 | -0.0733 | -0.0180 | -0.0287             | -0.0447             | -0.0552             | -0.0858            | -0.0958            |
|        | MSPE   | 0.1286  | 0.1297  | 0.1286  | 0.1265  | 0.1242  | 0.1242              | 0.1247              | 0.1253              | 0.1284             | 0.1298             |
| X3,(3) | Bias   | -0.1329 | -0.0031 | -0.0472 | -0.1085 | -0.0371 | -0.0526             | -0.0755             | -0.0906             | -0.1352            | -0.1498            |
|        | MSPE   | 0.2749  | 0.2684  | 0.2667  | 0.2617  | 0.2565  | 0.2568              | 0.2582              | 0.2598              | 0.2670             | 0.2703             |
| X5,(1) | Bias   | -0.0089 | -0.0005 | -0.0513 | -0.0673 | -0.0084 | -0.0133             | -0.0204             | -0.0249             | -0.0378            | -0.0418            |
|        | MSPE   | 0.0479  | 0.0480  | 0.0497  | 0.0508  | 0.0472  | 0.0473              | 0.0474              | 0.0475              | 0.0482             | 0.0485             |
| X5,(2) | Bias   | -0.0540 | -0.0142 | -0.0540 | -0.0889 | -0.0310 | -0.0393             | -0.0515             | -0.0594             | -0.0824            | -0.0898            |
|        | MSPE   | 0.1174  | 0.1168  | 0.1174  | 0.1174  | 0.1129  | 0.1132              | 0.1138              | 0.1144              | 0.1169             | 0.1179             |
| X5,(3) | Bias   | -0.1486 | -0.0218 | -0.0682 | -0.1239 | -0.0506 | -0.0638             | -0.0834             | -0.0962             | -0.1335            | -0.1457            |
|        | MSPE   | 0.2422  | 0.2303  | 0.2308  | 0.2291  | 0.2210  | 0.2218              | 0.2235              | 0.2250              | 0.2315             | 0.2342             |
| X8,(1) | Bias   | -0.0111 | -0.0060 | -0.0374 | -0.0450 | -0.0096 | -0.0107             | -0.0124             | -0.0135             | -0.0166            | -0.0175            |
|        | MSPE   | 0.0160  | 0.0160  | 0.0169  | 0.0172  | 0.0157  | 0.0157              | 0.0157              | 0.0158              | 0.0158             | 0.0158             |
| X8,(2) | Bias   | -0.0200 | -0.0065 | -0.0355 | -0.0541 | -0.0144 | -0.0167             | -0.0201             | -0.0223             | -0.0287            | -0.0307            |
|        | MSPE   | 0.0306  | 0.0305  | 0.0311  | 0.0315  | 0.0296  | 0.0296              | 0.0297              | 0.0297              | 0.0299             | 0.0300             |
| X8,(3) | Bias   | -0.0459 | -0.0185 | -0.0459 | -0.0753 | -0.0311 | -0.0347             | -0.0400             | -0.0435             | -0.0535            | -0.0567            |
|        | MSPE   | 0.0561  | 0.0554  | 0.0561  | 0.0568  | 0.0534  | 0.0535              | 0.0538              | 0.0540              | 0.0547             | 0.0550             |
| X8,(4) | Bias   | -0.0606 | -0.0061 | -0.0355 | -0.0780 | -0.0257 | -0.0312             | -0.0393             | -0.0446             | -0.0599            | -0.0649            |
|        | MSPE   | 0.0995  | 0.0992  | 0.0987  | 0.0964  | 0.0936  | 0.0936              | 0.0938              | 0.0940              | 0.0949             | 0.0953             |
| X8,(5) | Bias   | -0.1540 | -0.0230 | -0.0667 | -0.1252 | -0.0525 | -0.0619             | -0.0756             | -0.0846             | -0.1107            | -0.1191            |
|        | MSPE   | 0.2107  | 0.1990  | 0.1990  | 0.1952  | 0.1875  | 0.1880              | 0.1889              | 0.1897              | 0.1931             | 0.1945             |

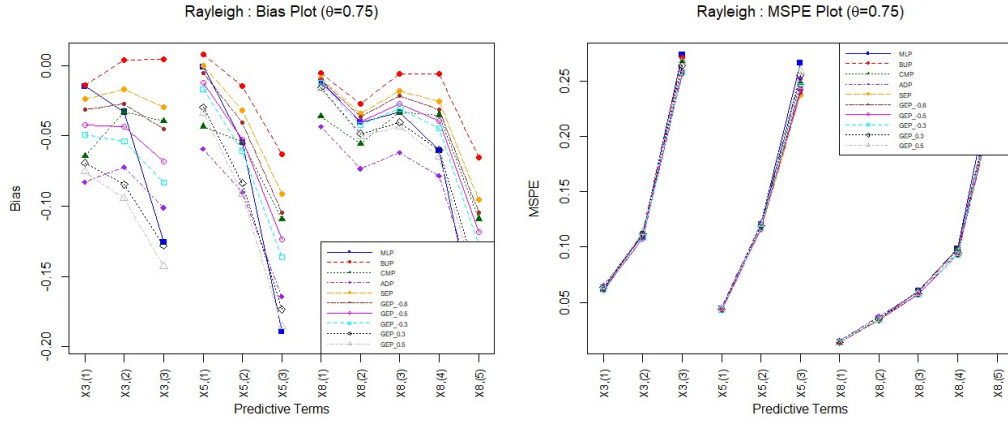


Figure 9: Bias and MSPE plots for Rayleigh Distribution ( $\theta = 0.75$ )

Table 12: Bias and MSPE Values for Rayleigh Distribution( $\theta = 1$ )

| Term   | Metric | MLP      | BUP      | CMP      | ADP      | SEP      | GEP <sub>-0.8</sub> | GEP <sub>-0.5</sub> | GEP <sub>-0.3</sub> | GEP <sub>0.3</sub> | GEP <sub>0.5</sub> |
|--------|--------|----------|----------|----------|----------|----------|---------------------|---------------------|---------------------|--------------------|--------------------|
| X3,(1) | Bias   | -0.01069 | -0.01006 | -0.05346 | -0.06422 | -0.01210 | -0.01883            | -0.02860            | -0.03492            | -0.05266           | -0.05823           |
|        | MSPE   | 0.04778  | 0.04779  | 0.04946  | 0.04955  | 0.04657  | 0.04668             | 0.04701             | 0.04732             | 0.04871            | 0.04928            |
| X3,(2) | Bias   | -0.04105 | -0.00963 | -0.04104 | -0.06363 | -0.01535 | -0.02489            | -0.03897            | -0.04821            | -0.07525           | -0.08402           |
|        | MSPE   | 0.09501  | 0.09490  | 0.09500  | 0.09263  | 0.09061  | 0.09066             | 0.09109             | 0.09160             | 0.09412            | 0.09526            |
| X3,(3) | Bias   | -0.12940 | -0.01769 | -0.05558 | -0.09114 | -0.02893 | -0.04253            | -0.06273            | -0.07606            | -0.11544           | -0.12833           |
|        | MSPE   | 0.19533  | 0.18646  | 0.18658  | 0.18095  | 0.17683  | 0.17710             | 0.17822             | 0.17941             | 0.18509            | 0.18764            |
| X5,(1) | Bias   | -0.01462 | -0.00724 | -0.05157 | -0.06136 | -0.00917 | -0.01360            | -0.01999            | -0.02409            | -0.03566           | -0.03928           |
|        | MSPE   | 0.04107  | 0.04108  | 0.04272  | 0.04299  | 0.04016  | 0.04020             | 0.04032             | 0.04045             | 0.04101            | 0.04124            |
| X5,(2) | Bias   | -0.05856 | -0.02399 | -0.05855 | -0.07936 | -0.02842 | -0.03586            | -0.04672            | -0.05381            | -0.07432           | -0.08090           |
|        | MSPE   | 0.09438  | 0.09306  | 0.09438  | 0.09333  | 0.08958  | 0.08981             | 0.09039             | 0.09090             | 0.09294            | 0.09379            |
| X5,(3) | Bias   | -0.12953 | -0.01846 | -0.05912 | -0.09298 | -0.02804 | -0.03986            | -0.05731            | -0.06876            | -0.10222           | -0.11307           |
|        | MSPE   | 0.19875  | 0.18979  | 0.19008  | 0.18510  | 0.18078  | 0.18101             | 0.18187             | 0.18277             | 0.18698            | 0.18884            |
| X8,(1) | Bias   | -0.00926 | -0.00489 | -0.03209 | -0.03678 | -0.00531 | -0.00635            | -0.00786            | -0.00884            | -0.01166           | -0.01256           |
|        | MSPE   | 0.01096  | 0.01095  | 0.01166  | 0.01178  | 0.01070  | 0.01070             | 0.01072             | 0.01073             | 0.01077            | 0.01079            |
| X8,(2) | Bias   | -0.03086 | -0.01931 | -0.04412 | -0.05482 | -0.02012 | -0.02220            | -0.02523            | -0.02720            | -0.03290           | -0.03472           |
|        | MSPE   | 0.02838  | 0.02808  | 0.02905  | 0.02901  | 0.02703  | 0.02709             | 0.02719             | 0.02726             | 0.02752            | 0.02762            |
| X8,(3) | Bias   | -0.03455 | -0.01087 | -0.03455 | -0.05189 | -0.01303 | -0.01628            | -0.02104            | -0.02415            | -0.03316           | -0.03606           |
|        | MSPE   | 0.04248  | 0.04230  | 0.04248  | 0.04113  | 0.03970  | 0.03972             | 0.03978             | 0.03985             | 0.04016            | 0.04030            |
| X8,(4) | Bias   | -0.06651 | -0.01930 | -0.04485 | -0.07026 | -0.02415 | -0.02905            | -0.03625            | -0.04095            | -0.05462           | -0.05903           |
|        | MSPE   | 0.08383  | 0.08225  | 0.08255  | 0.07969  | 0.07722  | 0.07730             | 0.07750             | 0.07769             | 0.07850            | 0.07885            |
| X8,(5) | Bias   | -0.17233 | -0.05845 | -0.09650 | -0.13080 | -0.06671 | -0.07503            | -0.08726            | -0.09525            | -0.11846           | -0.12596           |
|        | MSPE   | 0.19176  | 0.17348  | 0.17659  | 0.17291  | 0.16385  | 0.16462             | 0.16601             | 0.16708             | 0.17095            | 0.17243            |

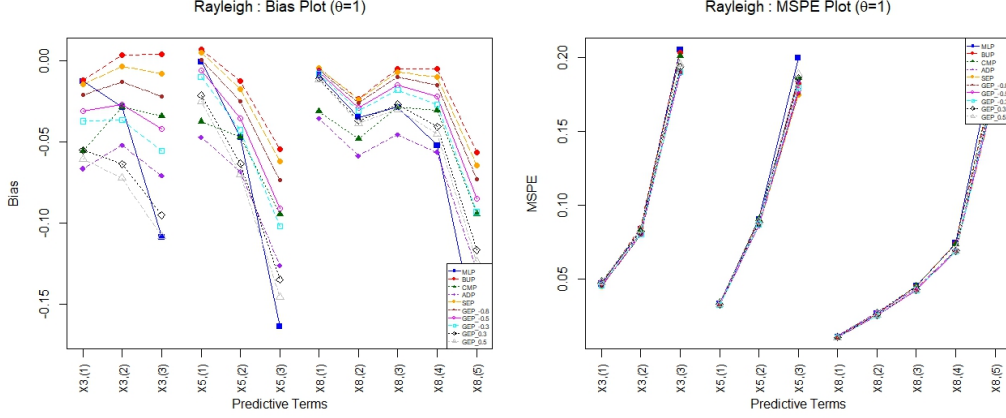


Figure 10: Bias and MSPE plots for Rayleigh Distribution ( $\theta = 1$ )

Table 13: Bias and MSPE for Rayleigh Distribution( $\theta = 2$ )

| Term   | Metric | MLP       | BUP        | CMP       | ADP       | SEP        | GEP <sub>-0.8</sub> | GEP <sub>-0.5</sub> | GEP <sub>-0.3</sub> | GEP <sub>0.3</sub> | GEP <sub>0.5</sub> |
|--------|--------|-----------|------------|-----------|-----------|------------|---------------------|---------------------|---------------------|--------------------|--------------------|
| X3,(1) | Bias   | -0.006126 | -0.0055364 | -0.036721 | -0.030827 | 0.0085324  | 0.003227            | -0.004477           | -0.009441           | -0.023494          | -0.0278946         |
|        | MSPE   | 0.025071  | 0.0251077  | 0.025934  | 0.024882  | 0.0243971  | 0.024308            | 0.024286            | 0.024340            | 0.024781           | 0.0250072          |
| X3,(2) | Bias   | -0.037525 | -0.0151299 | -0.037518 | -0.025715 | 0.0102698  | 0.002881            | -0.008063           | -0.015261           | -0.036358          | -0.0432138         |
|        | MSPE   | 0.047884  | 0.0473422  | 0.047884  | 0.044584  | 0.0447029  | 0.044470            | 0.044333            | 0.044377            | 0.045124           | 0.0455659          |
| X3,(3) | Bias   | -0.096684 | -0.0166643 | -0.043803 | -0.025325 | 0.0213748  | 0.010957            | -0.004532           | -0.014846           | -0.045008          | -0.0549297         |
|        | MSPE   | 0.102072  | 0.0967495  | 0.097077  | 0.088968  | 0.0902649  | 0.089620            | 0.089073            | 0.089006            | 0.089965           | 0.0906972          |
| X5,(1) | Bias   | -0.019330 | -0.0142341 | -0.045007 | -0.039192 | -0.0004798 | -0.004012           | -0.009115           | -0.012388           | -0.021615          | -0.0244998         |
|        | MSPE   | 0.022224  | 0.0221804  | 0.023484  | 0.022327  | 0.0211977  | 0.021203            | 0.021257            | 0.021321            | 0.021620           | 0.0217500          |
| X5,(2) | Bias   | -0.015538 | 0.0089735  | -0.015533 | -0.004318 | 0.0335929  | 0.027765            | 0.019184            | 0.013584            | -0.002621          | -0.0078263         |
|        | MSPE   | 0.037687  | 0.0382658  | 0.037687  | 0.035045  | 0.0369112  | 0.036480            | 0.035970            | 0.035722            | 0.035368           | 0.0353701          |
| X5,(3) | Bias   | -0.073365 | 0.0062800  | -0.022883 | -0.005087 | 0.0436465  | 0.034487            | 0.020938            | 0.011828            | -0.013960          | -0.0224158         |
|        | MSPE   | 0.093417  | 0.0921675  | 0.091088  | 0.083370  | 0.0870043  | 0.086013            | 0.084863            | 0.084296            | 0.083644           | 0.0837184          |
| X8,(1) | Bias   | -0.003690 | -0.0006335 | -0.019617 | -0.017086 | 0.0069309  | 0.006069            | 0.004815            | 0.004002            | 0.001675           | 0.0009336          |
|        | MSPE   | 0.005305  | 0.0053179  | 0.005551  | 0.005318  | 0.0051791  | 0.005166            | 0.005149            | 0.005140            | 0.005121           | 0.0051174          |
| X8,(2) | Bias   | -0.008543 | -0.0003451 | -0.017933 | -0.012152 | 0.0142197  | 0.012523            | 0.010041            | 0.008428            | 0.003774           | 0.0022822          |
|        | MSPE   | 0.012548  | 0.0126016  | 0.012643  | 0.011795  | 0.0121111  | 0.012051            | 0.011975            | 0.011931            | 0.011837           | 0.0118157          |
| X8,(3) | Bias   | -0.021478 | -0.0046768 | -0.021476 | -0.011897 | 0.0173932  | 0.014757            | 0.010896            | 0.008370            | 0.001061           | -0.0012878         |
|        | MSPE   | 0.020809  | 0.0207999  | 0.020810  | 0.018693  | 0.0193194  | 0.019204            | 0.019062            | 0.018985            | 0.018836           | 0.0188109          |
| X8,(4) | Bias   | -0.047705 | -0.0147985 | -0.032592 | -0.016675 | 0.0173475  | 0.013489            | 0.007793            | 0.004099            | -0.006739          | -0.0102318         |
|        | MSPE   | 0.037203  | 0.0364347  | 0.036571  | 0.031626  | 0.0325155  | 0.032329            | 0.032105            | 0.031990            | 0.031839           | 0.0318390          |
| X8,(5) | Bias   | -0.090501 | -0.0089592 | -0.036203 | -0.015788 | 0.0323995  | 0.025880            | 0.016283            | 0.010008            | -0.008240          | -0.0141348         |
|        | MSPE   | 0.080785  | 0.0768958  | 0.076693  | 0.067978  | 0.0704823  | 0.069901            | 0.069202            | 0.068846            | 0.068268           | 0.0682271          |

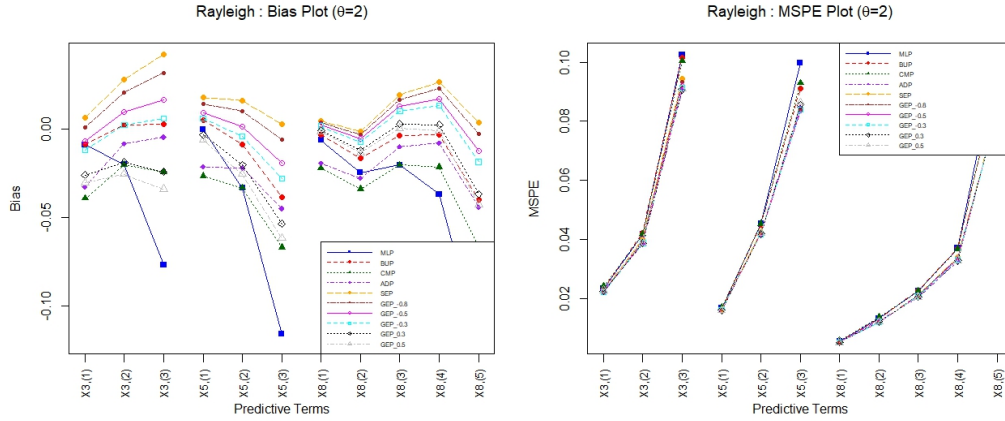


Figure 11: Bias and MSPE plots for Rayleigh Distribution ( $\theta = 2$ )

## References

- Asgharzadeh, A. and Valiollahi, R. (2009), Inference for the proportional hazards family under progressive Type-II censoring. *JIRSS.*, 1-2, 35–53.
- Asgharzadeh, A. and Valiollahi, R. (2010), Prediction intervals for proportional hazard rate models based on progressively Type II censored samples. *Communications of the Korean Statistical Society*, 17(1), 99–106.
- Ahmadi, J., Jafari Jozani, M., Marchand, E., and Parsian, A. (2009), Prediction of  $k$ -records from a general class of distributions under balanced type loss functions. *Metrika.*, 70, 19–33.
- Ahmadi, J., Jafari Jozani, M., Marchand, E., and Parsian, A. (2009), Bayesian estimation based on  $k$ -record data from a general class of distributions under balanced type loss functions. *Journal of Statistical Planning and Inference.*, 139, 1180–1189.
- Balakrishnan, N. and Aggarwala, R. (2000), *Progressive Censoring: Theory, Methods and Applications*. Boston: Birkhäuser.
- Balakrishnan N. and Lin C. T. (2002), Exact linear inference and prediction for exponential distributions based on general progressively Type-II censored

- samples. *Journal of Statistics Computation and Simulation.*, 72, 677–686.
- Basak, I., Basak, P., and Balakrishnan, N. (2006), On some predictors of times to failure of censored items in progressively censored samples. *Computational Statistics and Data Analysis.*, 50, 1313–1337.
- Basak, P. and Balakrishnan, N. (2003), Maximum likelihood prediction of future record statistic. *Mathematical and statistical methods in reliability*. In: Lindquist BH, Doksum KA (eds) Series on quality, reliability and engineering statistics. World Scientific Publishing, Singapore, 159–175.
- Cox, D. R. (1972), Regression models and life tables (with discussion). *Journal of the Royal Statistical Society, Series B.*, 34, 187–220.
- Dey, D. K. (1999), The discussion of paper of R.L. Smith. In: Berger, J.O., Bernardo, J.M., Dawid, A.P., Smith, A.F.M. (Eds), *Bayesian Statistics*, vol. 6. Oxford University Press, Oxford, 604–606.
- Kaminsky, K. S. and Rhodin, L. S. (1985), Maximum likelihood prediction. *Annals of the Institute of Statistical Mathematics*, 37, 507–517.
- Lawless, J. F. (1982), *Statistical Models and Methods for Life Time Data*. New York: John Wiley and Sons.
- Marshall, A. W. and Olkin, O. (2007), *Life Distributions*. New York: Springer.
- Nayak, T. K. (2000), On best unbiased prediction and its relationships to unbiased estimation. *Journal of Statistical Planning Inference*, 84, 171–189.
- Raqab, M. Z., Asgharzadeh, A., and Valiollahi, R. (2010), Prediction for Pareto distribution based on progressively Type-II censored samples. *Computational Statistics and Data Analysis*, 54, 1732–1743.
- Raqab, M. Z. and Madi, M. T. (2011), Inference for the generalized Rayleigh distribution based on progressively censored data. *Journal of Statistical Planning and Inference*, 141, 3312–3322.
- Raqab, M. Z. and Nagaraja, H. N. (1995), On some predictors of future order statistics. *Metron*, 53(12), 185–204.
- Soliman, A. A. (2005), Estimation of parameters of life from progressively

censored data using Burr-XII model. *IEEE Transactions on Reliability*, 54(1), 34–42.

Viveros, R. and Balakrishnan, N. (1994), Interval estimation of life characteristics from progressively censored data. *Technometrics*, 36, 84–91.

Zellner, A. (1986), Bayesian estimation and prediction using asymmetric loss function. *Journal of the American Statistical Association*, 81, 446–451.

## Appendix :

### Markovian Property under Progressive Type-II Censoring

Let  $X_1, X_2, \dots, X_m$  be progressively Type-II right censored order statistics from a continuous distribution with cdf  $F(\cdot)$  and pdf  $f(\cdot)$ , under censoring scheme  $(R_1, R_2, \dots, R_m)$ .

Let  $X_{i,(s)}$  denote the  $s$ -th order statistic among the  $R_i$  units removed at stage  $i$ .

#### Claim

$$f_{X_{i,(s)}|X}(y | x_1, \dots, x_m) = f_{X_{i,(s)}|X_i}(y | x_i) \quad (45)$$

#### Proof

At the  $i$ -th stage, immediately after observing the failure time  $X_i = x_i$ ,  $R_i$  surviving units are randomly removed from the remaining units.

Given  $X_i = x_i$ , all remaining units must have lifetimes exceeding  $x_i$ . Therefore, the lifetimes of the removed units follow the conditional (left-truncated) distribution:

$$f_{Y|Y \geq x_i}(y) = \frac{f(y)}{1 - F(x_i)}, \quad y \geq x_i. \quad (46)$$

Let  $Y_1, Y_2, \dots, Y_{R_i}$  denote the lifetimes of the removed units at stage  $i$ . Then, conditional on  $X_i = x_i$ , these variables are independent and identically distributed with the above density.

Further, due to the progressive censoring mechanism, the removed units at stage  $i$  depend only on the current risk set and are independent of the past and future order statistics, given  $X_i$ . Hence,

$$\{Y_1, \dots, Y_{R_i}\} \perp\!\!\!\perp (X_j, j \neq i) \mid X_i. \quad (47)$$

Thus,

$$X_{i,(s)} \perp\!\!\!\perp (X_1, \dots, X_{i-1}, X_{i+1}, \dots, X_m) \mid X_i, \quad (48)$$

which implies

$$f_{X_{i,(s)}|X}(y \mid x_1, \dots, x_m) = f_{X_{i,(s)}|X_i}(y \mid x_i). \quad (49)$$

Now, since  $X_{i,(s)}$  is the  $s$ -th order statistic of  $R_i$  i.i.d. variables from the truncated distribution, its conditional density is given by the standard order statistic formula:

$$f_{X_{i,(s)}|X_i}(y \mid x_i) = s \binom{R_i}{s} \left[ \frac{F(y) - F(x_i)}{1 - F(x_i)} \right]^{s-1} \left[ \frac{1 - F(y)}{1 - F(x_i)} \right]^{R_i-s} \frac{f(y)}{1 - F(x_i)}, \quad y \geq x_i. \quad (50)$$

This completes the proof. ■

```
##Lindley

set.seed(123)

r <- c(0,0,3,0,3,0,0,5)
m <- length(r)
theta <- 1

w <- runif(m)

#sub function pdf
pdf <- function(z)
{
  (theta^2 / (1 + theta)) * (1 + z) * exp(-theta * z)
}
```



```

# subfunction cdf
cdf <- function(y)
{
  1 - ((1 + theta + theta*y) * exp(-theta*y)) / (1 + theta)
}

#initialization
v <- numeric(m)
u <- numeric(m)
x <- numeric(m)

#Step 1: v
for (i in 1:m)
{
  expr <- 0
  for (j in (m - i + 1):m) expr <- expr + r[j]
  v[i] <- w[i]^(1 / (i + expr))
}

##Step 2: u
for (i in 1:m)
{
  prodv <- 1
  for (j in (m - i + 1):m) prodv <- prodv * v[j]
  u[i] <- 1 - prodv
}

# Step 3: inversion
for (i in 1:m)
{
  initial <- 0.5
  absdiff <- 1

  while(absdiff > 1e-6)
  {
    f_val <- cdf(initial) - u[i]
    f_der <- pdf(initial)

```

```

    if(f_der == 0) break

    temp <- initial - f_val / f_der
    absdiff <- abs(temp - initial)
    initial <- temp
  }

  x[i] <- initial
}

## Sample generation complete
matrix(x,nrow=2 )

```

```

## Exponential Sample Generation
set.seed(123)

r <- c(0,0,3,0,3,0,0,5)
m <- length(r)
theta <- 1 # Rate parameter

w <- runif(m)

# Initialization
v <- numeric(m)
u <- numeric(m)
x <- numeric(m)

# Step 1: Generate v
for (i in 1:m) {
  expr <- 0
  for (j in (m - i + 1):m) expr <- expr + r[j]
  v[i] <- w[i]^(1 / (i + expr))
}

# Step 2: Generate u (uniform order statistics)
for (i in 1:m) {

```

```

    prodv <- 1
    for (j in (m - i + 1):m) prodv <- prodv * v[j]
    u[i] <- 1 - prodv
  }

  # Step 3: Direct Inversion
  for (i in 1:m) {
    # Inverse CDF:  $x = - (1/\theta) * \log(1 - u)$ 
    x[i] <- - (1/theta) * log(1 - u[i])
  }

  ## Sample generation complete
  cat("Exponential Samples:\n")
  matrix(x, nrow=2 )

```

```

## Rayleigh Sample Generation
set.seed(123)

r <- c(0,0,3,0,3,0,0,5)
m <- length(r)
theta <- 1 # Parameter (where baseline is  $\exp(-\theta * x^2)$ )

w <- runif(m)

# Initialization
v <- numeric(m)
u <- numeric(m)
x <- numeric(m)

# Step 1: Generate v
for (i in 1:m) {
  expr <- 0
  for (j in (m - i + 1):m) expr <- expr + r[j]
  v[i] <- w[i]^(1 / (i + expr))
}

# Step 2: Generate u

```

```

for (i in 1:m) {
  prodv <- 1
  for (j in (m - i + 1):m) prodv <- prodv * v[j]
  u[i] <- 1 - prodv
}

# Step 3: Direct Inversion
for (i in 1:m) {
  # Inverse CDF:  $x = \sqrt{- (1/\theta) * \log(1 - u)}$ 
  x[i] <- sqrt(- (1/theta) * log(1 - u[i]))
}

## Sample generation complete
cat("Rayleigh Samples:\n")
matrix(x, nrow=2 )

```

```

# MASTER SIMULATION SCRIPT: EXPONENTIAL, RAYLEIGH, AND LINDLEY DISTRIBUTIONS

```

```

library(lamW) # Required for the Lindley distribution inverse CDF

```

```

# Global Configuration & Shared Plotting Parameters

```

```

groups <- list(1:3, 4:6, 7:11)
cols <- c("blue", "red", "darkgreen", "purple", "orange", "brown",
          "magenta", "cyan", "black", "gray")
pchs <- c(15, 16, 17, 18, 19, 20, 21, 22, 23, 24)

col_names <- c("MLP", "BUP", "CMP", "ADP", "SEP",
               "GEP_-0.8", "GEP_-0.5", "GEP_-0.3",
               "GEP_0.3", "GEP_0.5")

```

```

# CUSTOM PLOTTING FUNCTION

```

```

create_grouped_plot <- function(mat, main_title, y_label,
                                row_labels, leg_x, leg_y = NULL) {

```

```

plot(1:nrow(mat), mat[,1], type = "n",
     ylim = range(mat, na.rm = TRUE),
     xaxt = "n",
     xlab = "Predictive Terms",
     ylab = y_label,
     main = main_title)

axis(1, at = 1:nrow(mat),
     labels = row_labels,
     las = 2,
     cex.axis = 0.8)

for(col in 1:ncol(mat)) {
  for(g in groups) {
    lines(g, mat[g,col],
          col = cols[col],
          lty = col,
          type = "o",
          pch = pchs[col])
  }
}

legend(x = leg_x,
       y = leg_y,
       legend = col_names,
       col = cols,
       lty = 1:10,
       pch = pchs,
       cex = 0.5,
       bg = "white",
       x.intersp = 0.4)
}

# COMMON SETTINGS

set.seed(123)

```

```

N_sim      <- 500
alpha      <- 2
beta       <- 1
theta_true <- 0.75

r_scheme <- c(0,0,3,0,3,0,0,5)
m <- length(r_scheme)

terms_list <- list(
  c(3,1), c(3,2), c(3,3),
  c(5,1), c(5,2), c(5,3),
  c(8,1), c(8,2), c(8,3), c(8,4), c(8,5)
)

n_predictors <- length(col_names)

row_labels <- sapply(terms_list,
  function(z) paste0("X", z[1], ",(", z[2], ")"))

# GENERAL SIMULATION FUNCTION

run_simulation <- function(f0_bar, F0_inv, hazard_b, dist_name){

  bias_matrix <- matrix(0,
    nrow = length(terms_list),
    ncol = n_predictors)

  mspe_matrix <- matrix(0,
    nrow = length(terms_list),
    ncol = n_predictors)

  for(t_idx in 1:length(terms_list)) {

    i_idx <- terms_list[[t_idx]][1]
    s_idx <- terms_list[[t_idx]][2]

```

```

pred_errors <- matrix(NA,
                      nrow = N_sim,
                      ncol = n_predictors)

for(sim in 1:N_sim) {

  w <- runif(m)
  v <- numeric(m)

  for(j in 1:m) {
    rem_sum <- sum(r_scheme[(m-j+1):m])
    v[j] <- w[j]^(1/(j + rem_sum))
  }

  p_survival <- numeric(m)

  for(j in 1:m) {
    prodv <- 1
    for(k in (m-j+1):m)
      prodv <- prodv * v[k]

    p_survival[j] <- prodv
  }

  x_obs <- F0_inv(p_survival^(1/theta_true))

  xi <- x_obs[i_idx]
  Ri <- r_scheme[i_idx]

  T0 <- sum((r_scheme + 1) * (-log(f0_bar(x_obs))))
  theta_hat <- m/T0

  u_true <- rbeta(1, Ri-s_idx+1, s_idx)

  y_true <- F0_inv(
    f0_bar(xi) * u_true^(1/theta_true)
  )

```

```

estimates <- numeric(n_predictors)

# MLP

target_mlp <- function(y){
  log(f0_bar(y)) - log(f0_bar(xi)) +
    (1/theta_hat) *
    log((Ri+1)/(Ri-s_idx+1))
}

estimates[1] <- tryCatch(
  uniroot(target_mlp, c(xi, xi+40))$root,
  error = function(e) NA
)

# BUP

integrand_bup <- function(u){
  y_val <- f0_bar(xi) * u^(1/theta_hat)
  F0_inv(y_val) *
    dbeta(u, Ri-s_idx+1, s_idx)
}

estimates[2] <- tryCatch(
  integrate(Vectorize(integrand_bup), 0, 1)$value,
  error = function(e) NA
)

# CMP

medU <- qbeta(0.5, Ri-s_idx+1, s_idx)

estimates[3] <- F0_inv(
  f0_bar(xi) * medU^(T0/m)
)

```



```

# ADP

pred_survival <- function(y){

  sum_term <- 0

  for(j in 0:(s_idx-1)) {

    comb <- choose(s_idx-1, j)

    term <- ((-1)^j) /
      (Ri-s_idx+j+1)

    log_ratio <- log(f0_bar(y)) -
      log(f0_bar(xi))

    inner <- (
      1 - (Ri-s_idx+j+1) *
        log_ratio/(beta+T0)
    )^(-(m+alpha))

    sum_term <- sum_term +
      comb * term * inner
  }

  s_idx * choose(Ri, s_idx) * sum_term
}

target_adp <- function(y)
  pred_survival(y) - 0.5

estimates[4] <- tryCatch(
  uniroot(target_adp, c(xi, xi+40))$root,
  error = function(e) NA
)

```

```

# GEP ENGINE

gep_calc <- function(q_val){

  total_sum <- 0

  for(j in 0:(s_idx-1)) {

    comb <- choose(s_idx-1, j)
    sign <- (-1)^j

    A <- (Ri-s_idx+j+1)

    integrand_gep <- function(y){

      log_ratio <- log(f0_bar(y)) -
        log(f0_bar(xi))

      y^(-q_val) *
        hazard_b(y) *
        (1 - A*log_ratio/(beta+T0)
        )^(-(m+alpha+1))
    }

    I_val <- tryCatch(
      integrate(integrand_gep,
        lower = xi,
        upper = xi+50)$value,
      error = function(e) 0
    )

    total_sum <- total_sum +
      comb * sign * I_val
  }

  val <- s_idx * choose(Ri, s_idx) *
    (m+alpha)/(beta+T0) *

```

```

    total_sum

    return(val^(-1/q_val))
  }

  estimates[5] <- tryCatch(gep_calc(-1), error=function(e) NA)
  estimates[6] <- tryCatch(gep_calc(-0.8), error=function(e) NA)
  estimates[7] <- tryCatch(gep_calc(-0.5), error=function(e) NA)
  estimates[8] <- tryCatch(gep_calc(-0.3), error=function(e) NA)
  estimates[9] <- tryCatch(gep_calc(0.3), error=function(e) NA)
  estimates[10] <- tryCatch(gep_calc(0.5), error=function(e) NA)

  pred_errors[sim, ] <- estimates - y_true
}

bias_matrix[t_idx, ] <-
  colMeans(pred_errors, na.rm = TRUE)

mspe_matrix[t_idx, ] <-
  colMeans(pred_errors^2, na.rm = TRUE)
}

# FINAL TABLE

final_table <- data.frame()

for(i in 1:length(row_labels)) {

  bias_row <- data.frame(
    Term = row_labels[i],
    Metric = "Bias",
    t(bias_matrix[i, ])
  )

  mspe_row <- data.frame(
    Term = "",

```

```

    Metric = "MSPE",
    t(mspe_matrix[i, ])
  )

  colnames(bias_row)[3:12] <-
    colnames(mspe_row)[3:12] <-
    col_names

  final_table <- rbind(final_table,
                        bias_row,
                        mspe_row)
}

print(final_table,
      digits = 4,
      row.names = FALSE)

# PLOTS

X11()

par(mfrow = c(1,2),
    mar = c(7,4,4,2))

bias_title <- bquote(
  .(dist_name) ~ ":" ~
  "Bias Plot (" * theta * "=" * .(theta_true) * ")"
)

mspe_title <- bquote(
  .(dist_name) ~ ":" ~
  "MSPE Plot (" * theta * "=" * .(theta_true) * ")"
)

create_grouped_plot(
  bias_matrix,
  bias_title,

```

```

    "Bias",
    row_labels,
    leg_x = "bottomright",

)

create_grouped_plot(
  mspe_matrix,
  mspe_title,
  "MSPE",
  row_labels,
  leg_x = "topright"
)
}

# 1. EXPONENTIAL DISTRIBUTION

cat("\n--- Running Exponential Distribution Simulation ---\n")

f0_bar <- function(x) exp(-x)

F0_inv <- function(y)
  -log(pmin(pmax(y, 1e-10), 0.999999))

hazard_b <- function(y)
  rep(1, length(y))

run_simulation(f0_bar,
              F0_inv,
              hazard_b,
              "Exponential")

# 2. RAYLEIGH DISTRIBUTION

cat("\n--- Running Rayleigh Distribution Simulation ---\n")

```

```

f0_bar <- function(x) exp(-x^2)

F0_inv <- function(y)
  sqrt(-log(pmin(pmax(y,1e-10),0.999999)))

hazard_b <- function(y)
  2*y

run_simulation(f0_bar,
              F0_inv,
              hazard_b,
              "Rayleigh")

# 3. LINDLEY DISTRIBUTION

cat("\n--- Running Lindley Distribution Simulation ---\n")

f0_bar <- function(x)
  ((2+x)/2) * exp(-x)

F0_inv <- function(y){
  y <- pmin(pmax(y,1e-10),0.999999)

  -lambertWm1(-2*y*exp(-2)) - 2
}

hazard_b <- function(y)
  (1+y)/(2+y)

run_simulation(f0_bar,
              F0_inv,
              hazard_b,
              "Lindley")

```